



CMPT 413: Computational Linguistics
CMPT 713: Natural Language Processing

Angel Xuan Chang

2022-09-12

Teaching Staff

Instructor

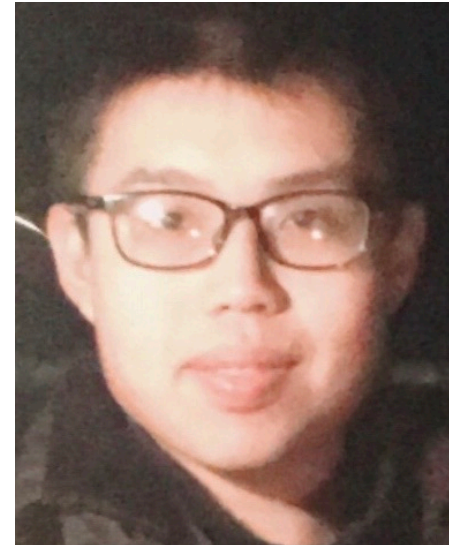


Angel Chang

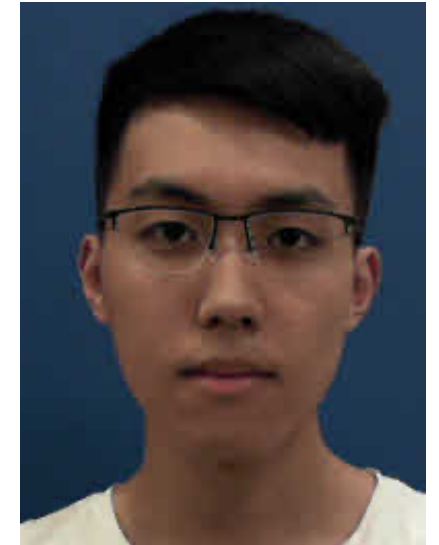
TAs



Han-Hung
Lee



Shawn
Jiang



Yiming
Zhang

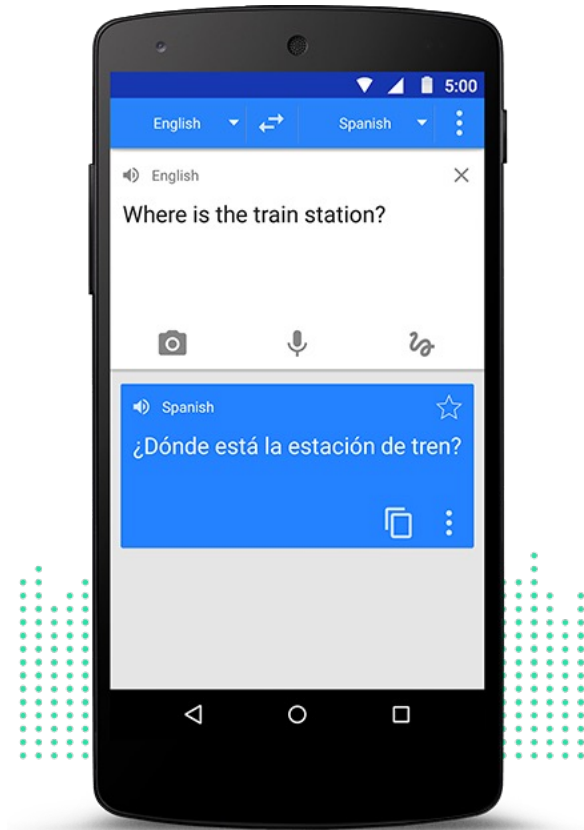
Resources

- Website: <https://angelxuanchang.github.io/nlp-class/>
- Coursys: <https://coursys.sfu.ca/2022fa-cmpt-413-x1/pages/>
- Lectures (in person)
 - Monday 2:30pm - 5:20pm AQ3159
 - Additional video lectures (optional)
- TA lead tutorials (optional)
 - 30 minute video
 - Interactive session: Wednesday 1:00pm – 1:30pm (over zoom)

Today

- Intro to NLP
 - What is language? NLP vs computational linguistics.
 - Brief history of NLP
 - Levels of representation in language
 - Key questions of NLP
- Course Logistics
- Intro into Language Modeling

NLP is everywhere



Google translate



Virtual assistants

Information finding



when were the first pyramids built



All

Images

News

Shopping

Videos

More

Settings

Tools

About 20,300,000 results (0.67 seconds)

2630 BC

Most were built as tombs for the country's pharaohs and their consorts during the Old and Middle Kingdom periods. The earliest known Egyptian pyramids are found at Saqqara, northwest of Memphis. The earliest among these is the Pyramid of Djoser (constructed **2630 BC**–2611 BC) which was built during the third dynasty.

[The First Pyramids Built - Timeline Index](#)

www.timelineindex.com › [content](#) › [view](#)

Text to image generation

 **OpenAI** DALL-E 2

a painting of a fox sitting in a field at sunrise in the style of Claude Monet



<https://openai.com/dall-e-2/>

 **Imagen**

A cute sloth holding a small treasure chest.
A bright golden glow is coming from the chest.

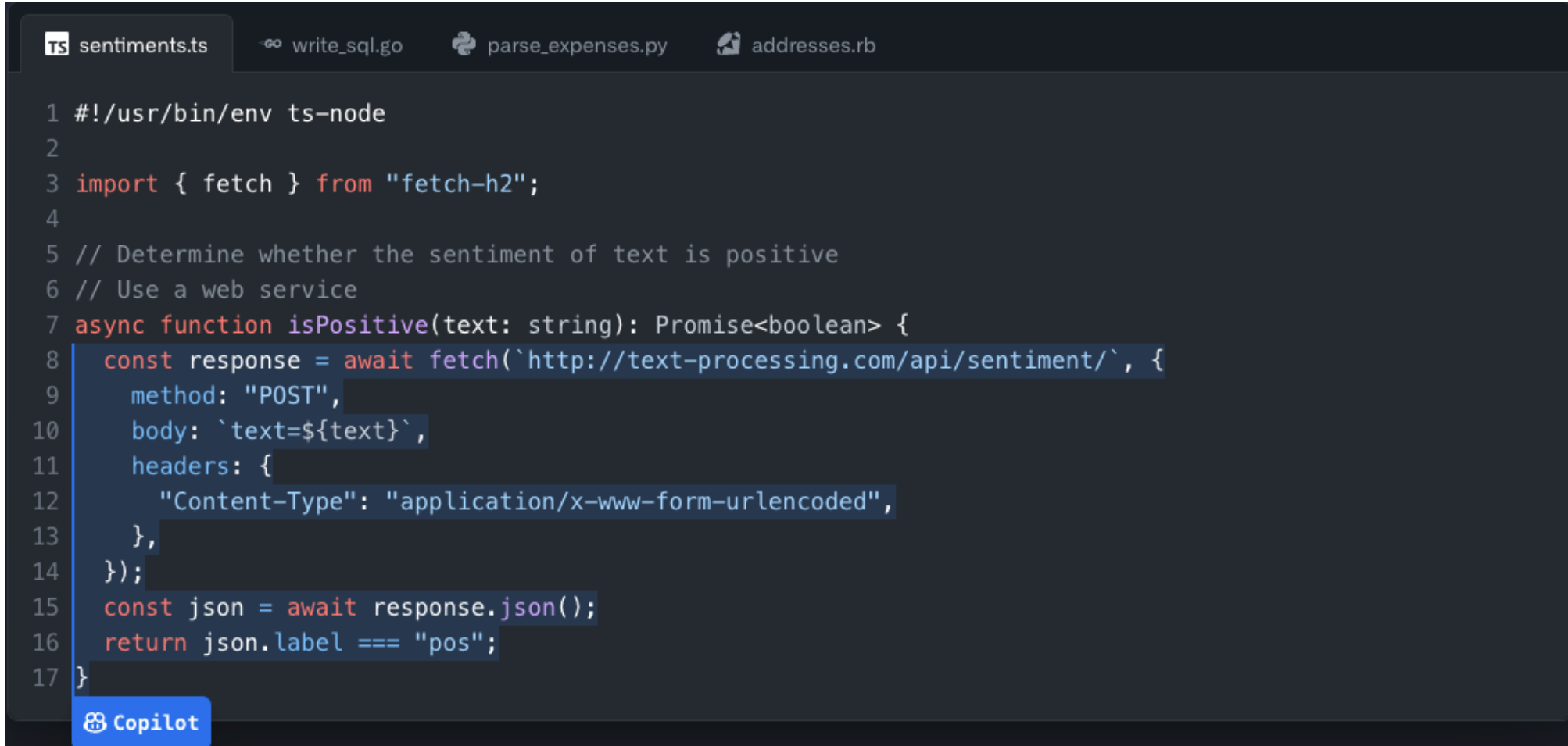


<https://imagen.research.google/>

Try it out yourself: <https://huggingface.co/spaces/stabilityai/stable-diffusion>,
<https://www.craiyon.com/>, <https://www.midjourney.com/home/>

Copilot

Generate code from text!



```
1 #!/usr/bin/env ts-node
2
3 import { fetch } from "fetch-h2";
4
5 // Determine whether the sentiment of text is positive
6 // Use a web service
7 async function isPositive(text: string): Promise<boolean> {
8   const response = await fetch(`http://text-processing.com/api/sentiment/`, {
9     method: "POST",
10    body: `text=${text}`,
11    headers: {
12      "Content-Type": "application/x-www-form-urlencoded",
13    },
14  });
15  const json = await response.json();
16  return json.label === "pos";
17 }
```

<https://github.com/features/copilot>

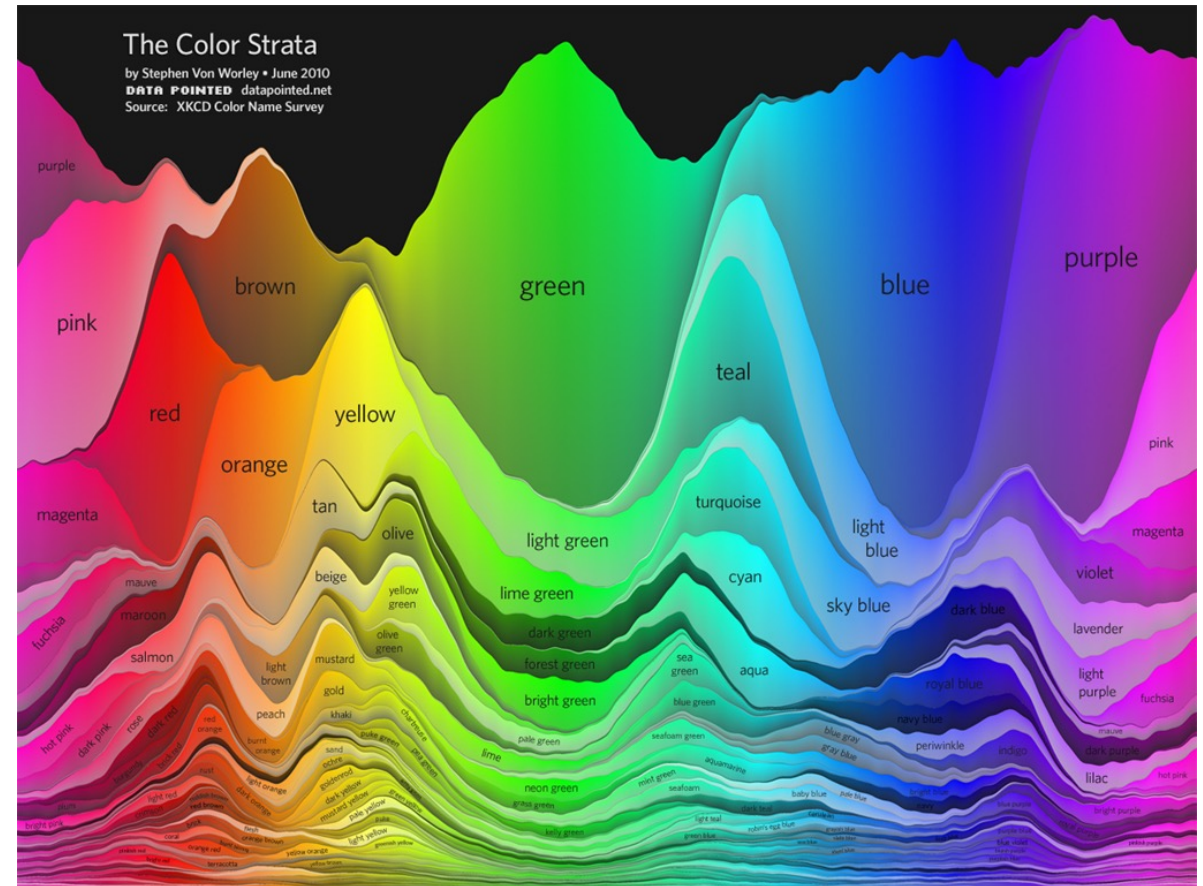
What is language?

- Language is used to communicate
 - Things, actions, abstract concepts



What is language?

- Language puts categories on the world
 - It discretizes a continuous space



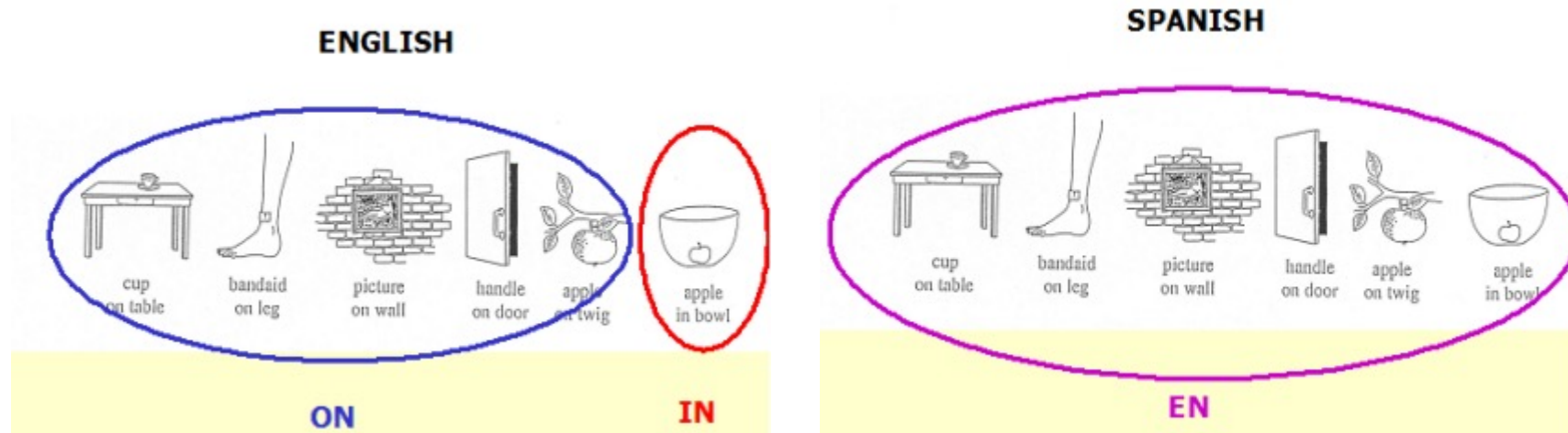
What is language?

- Language picks out what is salient and important
- What concepts do we have words for?
- Different languages have different discretization boundaries



What is language?

- Language picks out what is salient and important
- What concepts do we have words for?
- Different languages have different discretization boundaries



<http://pyersqr.org/classes/Ling731/Space2.htm>

Natural Language Processing



Building **useful system** to process language

Computational Linguistics

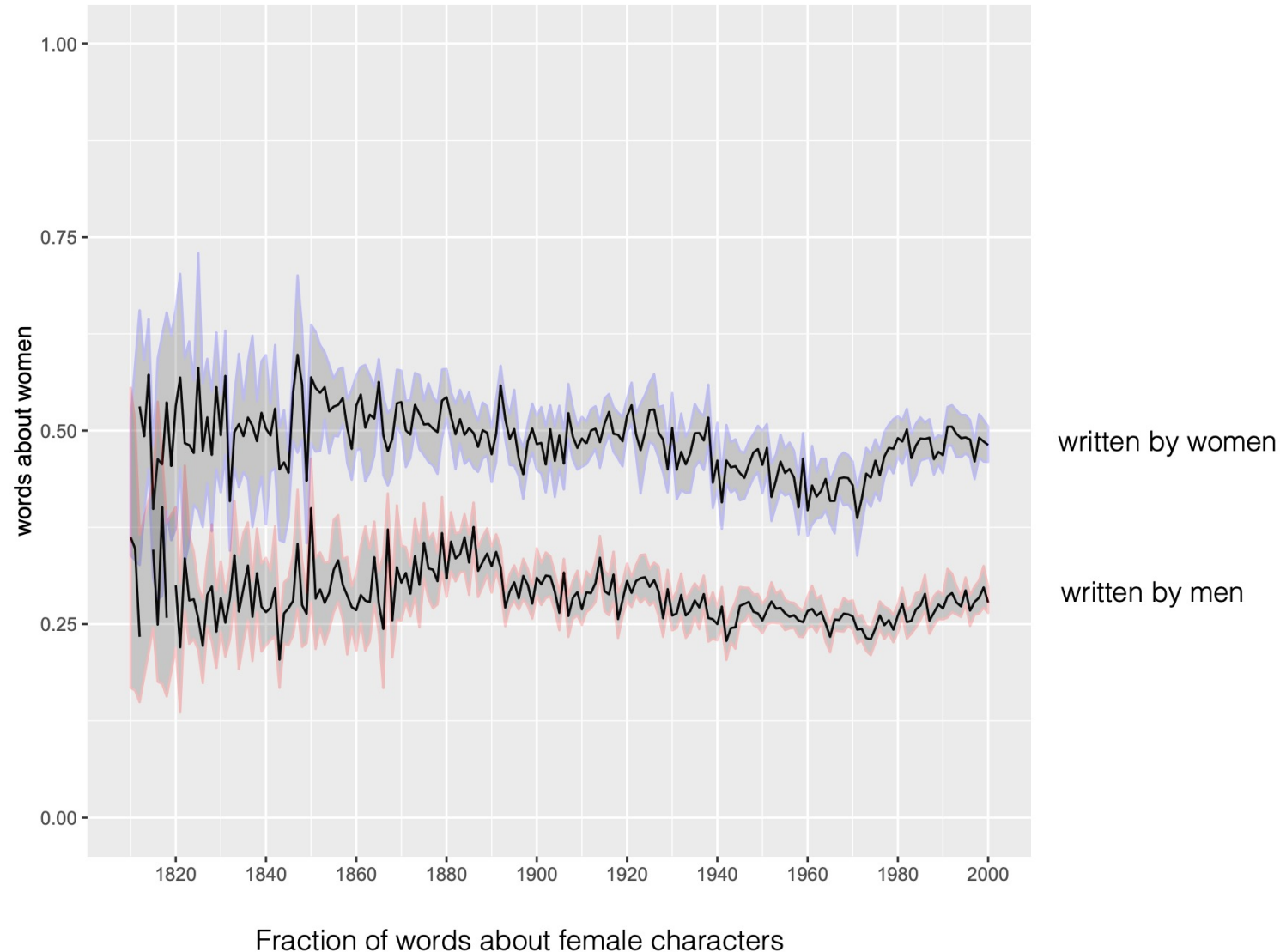


"I just had a heated conversation
with my computer."

(image credit: <https://www.enterrasolutions.com/blog/computational-linguistics-and-natural-language-processing/>)

Using computers to study **human** language

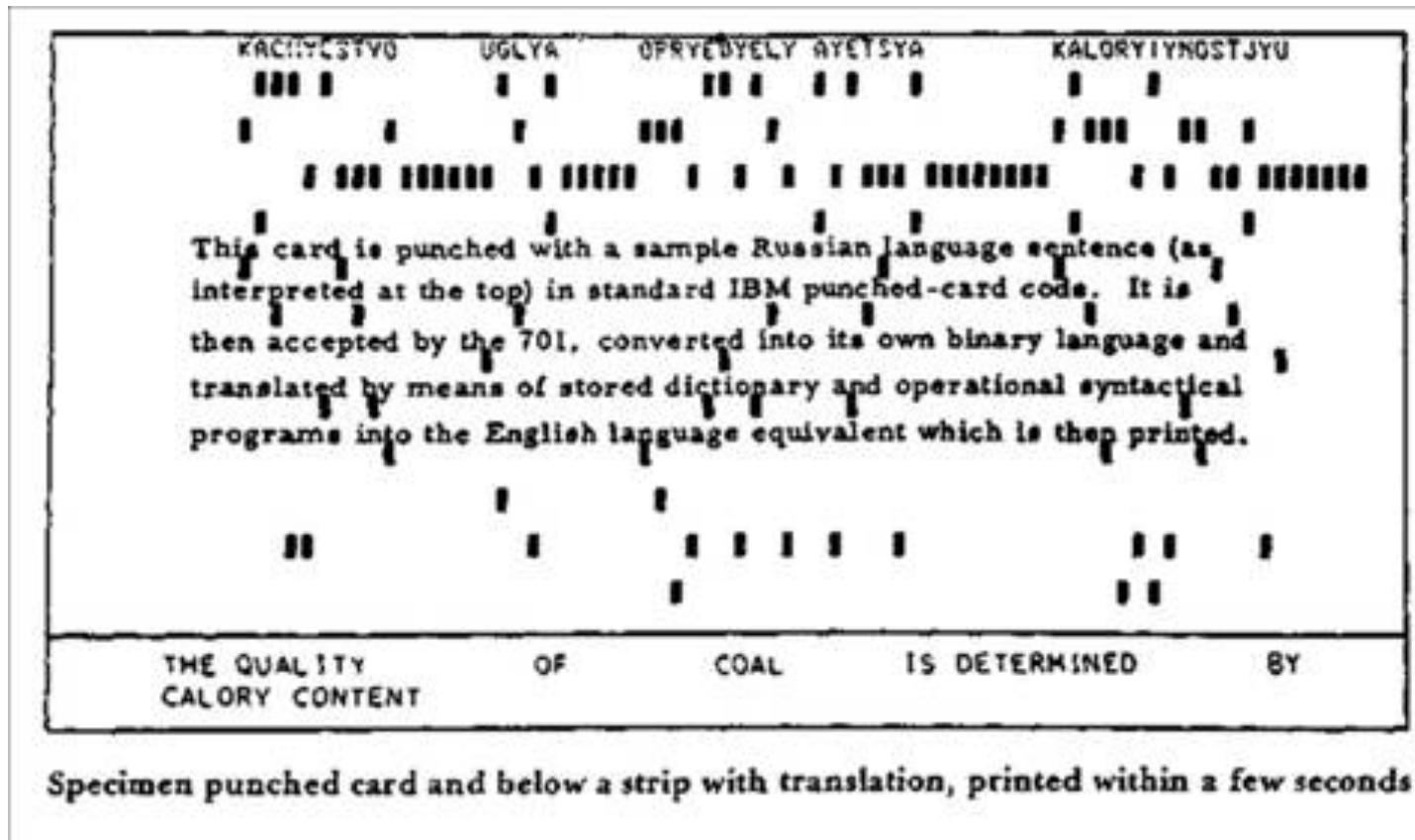
Analyzing word usage in literature



Ted Underwood, David Bamman, and Sabrina Lee (2018),
"The Transformation of Gender in English-Language Fiction," Cultural Analytics

Brief history of NLP

Beginnings



Georgetown-
IBM
experiment,
1954

“Within three or five years, machine translation will be a solved problem”

SHRDLU (Winograd, 1968)

Video of actual system: <https://www.youtube.com/watch?v=bo4RvYJYOzI>

Person: Pick up a big red block.

Computer: OK.

Person: Grasp the pyramid.

Computer: I don't understand which pyramid you mean.

Person (changing their mind): Find a block which is taller than the one you are holding and put it into the box.

Computer: By "it", I assume you mean the block which is taller than the one I am holding.

Computer: OK.

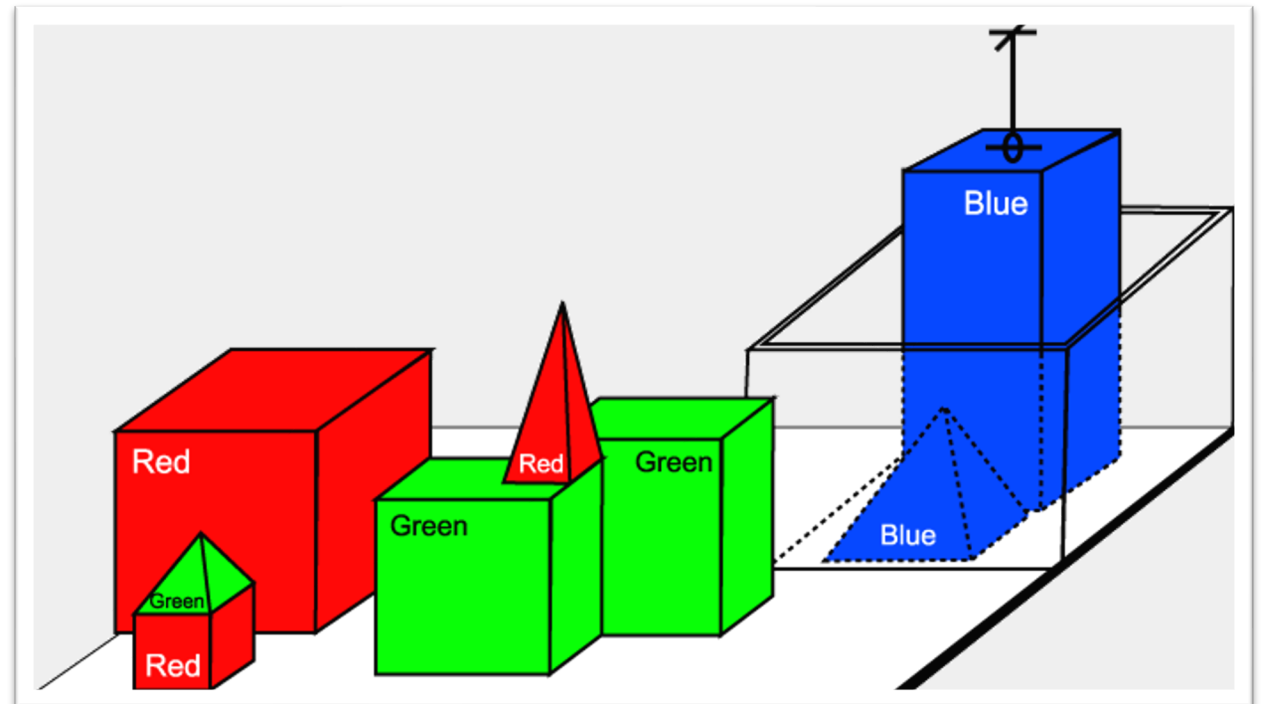
Person: What does the box contain?

Computer: The blue pyramid and the blue block.

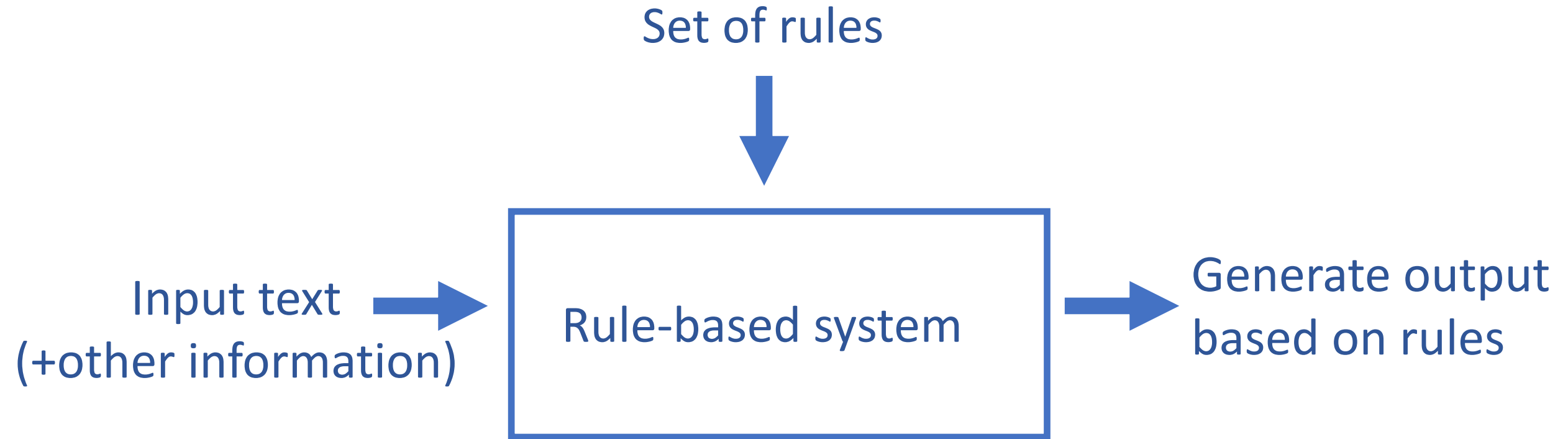
Person: What is the pyramid supported by?

Computer: The box.

Lots of rules!



Using rules for NLP



Match a pattern

- Regular expressions

 - *.ed => past tense verb

Contained keyword: delicious

Eliza (Weizenbaum, 1966)

Demo: <http://psych.fullerton.edu/mbirnbaum/psych101/Eliza.htm>

Men are all alike.

IN WHAT WAY

They're always bugging us about something or other.

CAN YOU THINK OF A SPECIFIC EXAMPLE

Well, my boyfriend made me come here.

YOUR BOYFRIEND MADE YOU COME HERE

He says I'm depressed much of the time.

I AM SORRY TO HEAR YOU ARE DEPRESSED

Rogarian psychologist:
reflect back what the patient said

- Set of ranked transformation **rules** based on **keywords**
- Apply rules based on matching words in sentence

Transformation rules

Keywords are ranked from specific to general

I know everybody laughed at me

- “I” is a very general keyword:

I: (I *) -> (You say you 2)

YOU SAY YOU KNOW EVERYBODY LAUGHED AT YOU

- “Everybody” is much more interesting (someone using universals like everybody/always is probably “referring to some quite specific event or person”)

WHO IN PARTICULAR ARE YOU THINKING OF?

- Implementation: keywords stored with their rank

Everybody 5 (*transformation rules*)

I 0 (*transformation rules*)

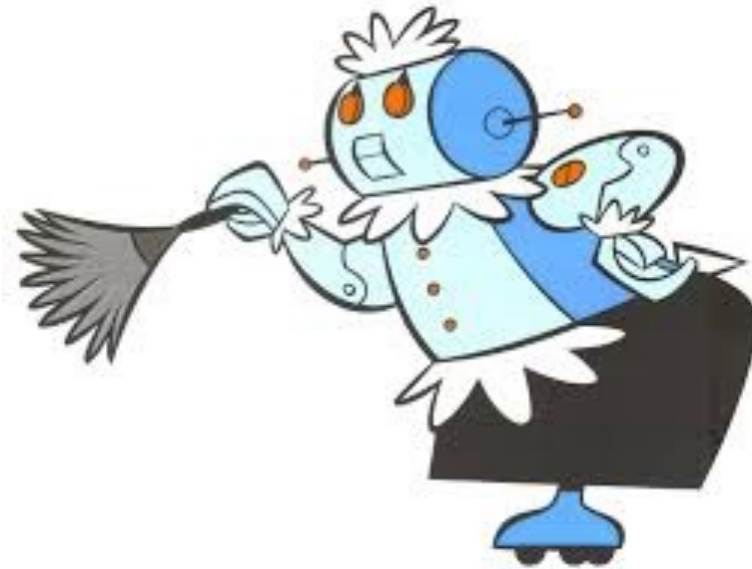
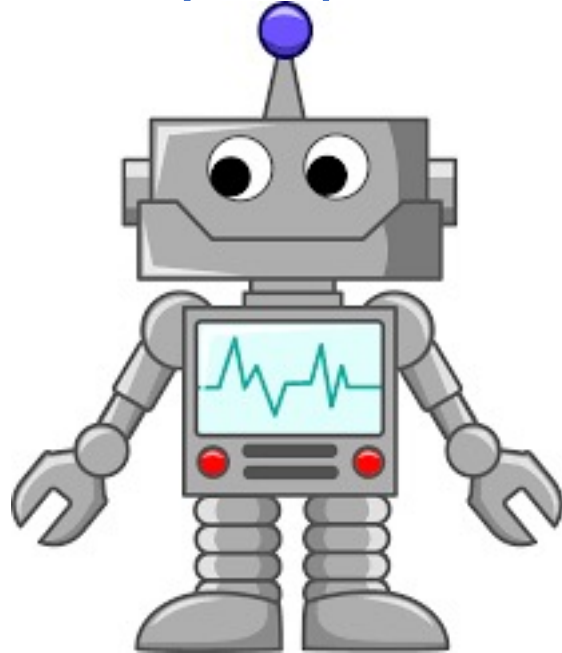
Backoff

Please go on

That's very interesting

I see

Where is my block stacking or
housekeeper robot that I can talk to?
(maybe we are getting close...)



Rosie from the Jetsons



Understanding
language is hard!

The Far Side - Gary Larson

Why is natural language processing hard?

Programming Languages

C, C++, Java, Python, ...

- Unambiguous
- Fixed
- Designed
- Learnable?
- Known simple semantics

Natural Languages

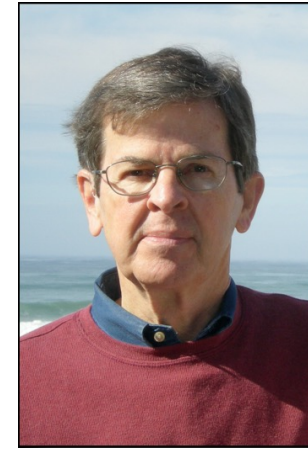
French, English, Korean, Chinese, Tagalog, ...

- Ambiguous
- Evolving
- Transmitted
- Learnable
- Complex semantics

Why is NLP hard?

Interpretation of language assumes a common basis of **world knowledge** and **context**

- **Ambiguous:**
 - “bank”, “bat”
 - “Milk Drinkers Turn to Powder”
- **Variation:** Many ways to say same thing
- **Context dependent:**
 - natural language is under-specified



Herb Clark

“bat’k”

Table



Counter

Context-dependence

“I put the bowl on the **table**”



“The numbers in the **table** don’t add up”

Table 8.2 Calculation of the χ^2 test on figures in table 8.1

Class (1)	Expected numbers		O - E		$(O-E)^2/E$	
	A (2)	B (3)	A (4)	B (5)	A (6)	B (7)
I	11.80	10.20	5.20	-5.20	2.292	2.651
II	24.67	21.33	0.33	-0.33	0.004	0.005
III	39.15	33.85	-0.15	0.15	0.001	0.001
IV	48.81	42.19	-6.81	6.81	0.950	1.009
V	30.57	26.43	1.43	-1.43	0.067	0.077
Total	30.57	134.00	0	0	3.314	3.833

$\chi^2 = 3.314 + 3.833 = 7.147$, d.f. = 4, $0.10 < P < 0.50$.

“I made her duck”

I cooked her duck (the waterfowl) for dinner

I cooked the duck (the waterfowl) belonging to her.

I created a model of a duck that she now has.

I caused her to lower her head

Some language humor (from real newspaper headlines)

Kids make nutritious snacks

Stolen painting found by tree

Miners refuse to work after death

Squad helps dog bite victim

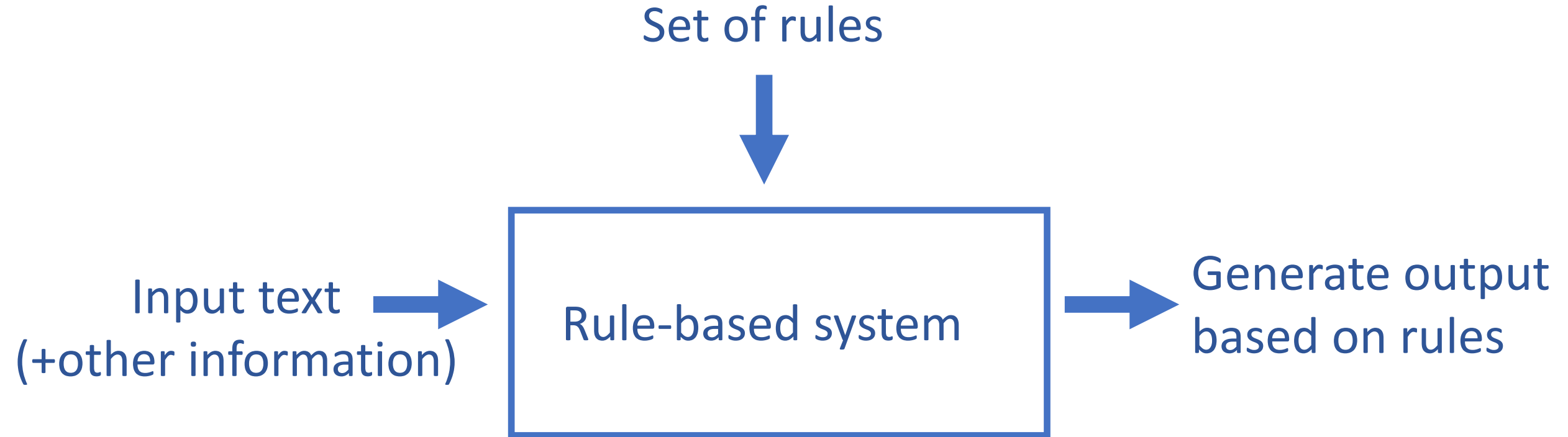
Killer sentenced to die for second time in 10 years

Lack of brains hinders research

Pooches and people digging newly opened park

Using rules for NLP

- Translation with rules started in the 1950s
- Eliza chatbot (1966)
- SHRDLU grounded language understanding (1968)



Rule-based Machine Translation

- Georgetown IBM translator
- Bilingual dictionary to map Russian words to their English counterparts
- But what if the word order is different? How to handle all the nuances of meaning?



1 minute video showing 1954 MT:

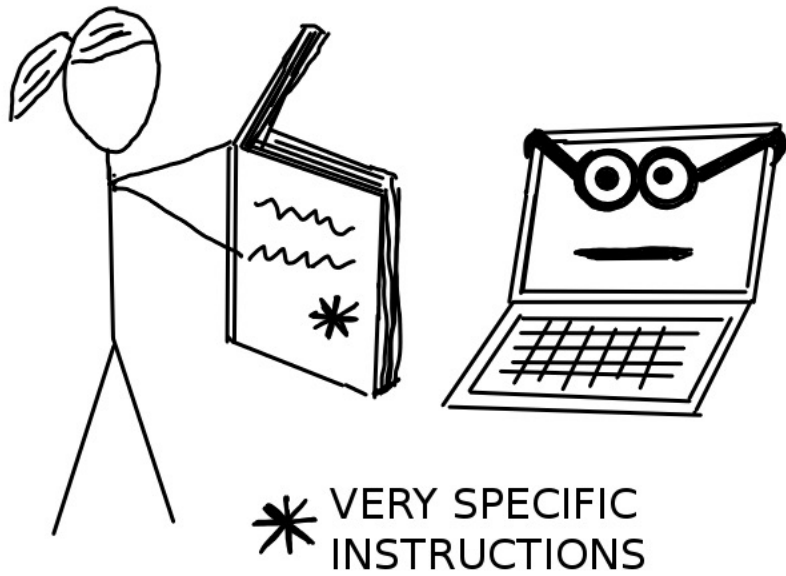
<https://youtu.be/K-HfpsHPmvw>

https://www.ibm.com/ibm/history/exhibits/701/701_translator.html

Coming up rules is hard!

Let's learn from data!

Without Machine Learning



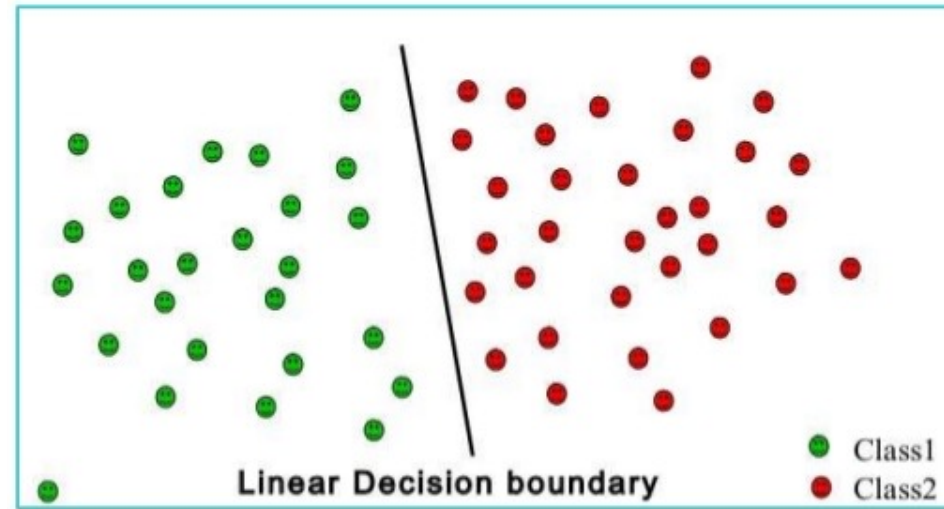
With Machine Learning



<https://christophm.github.io/interpretable-ml-book/terminology.html>

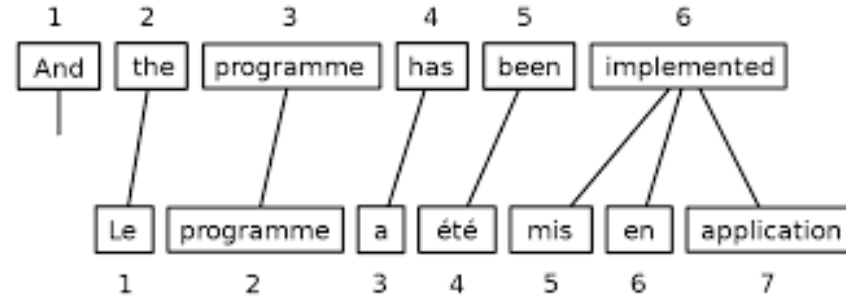
Rise of statistical learning

(late) 1980s to 2000s: Statistical MT, 2000s-2014: Statistical Phrase-Based MT

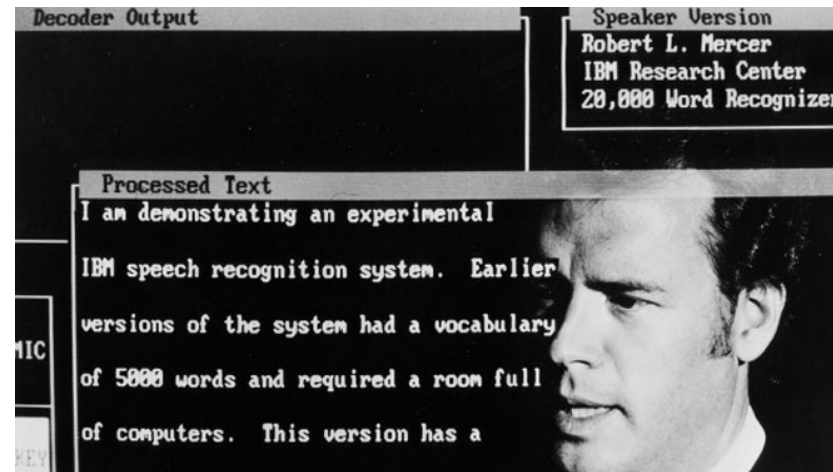


- Use of machine learning techniques in NLP
- Increase in computational capabilities
- Availability of electronic corpora
- Data-driven evaluation of models

Rise of statistical learning



IBM Models
for translation



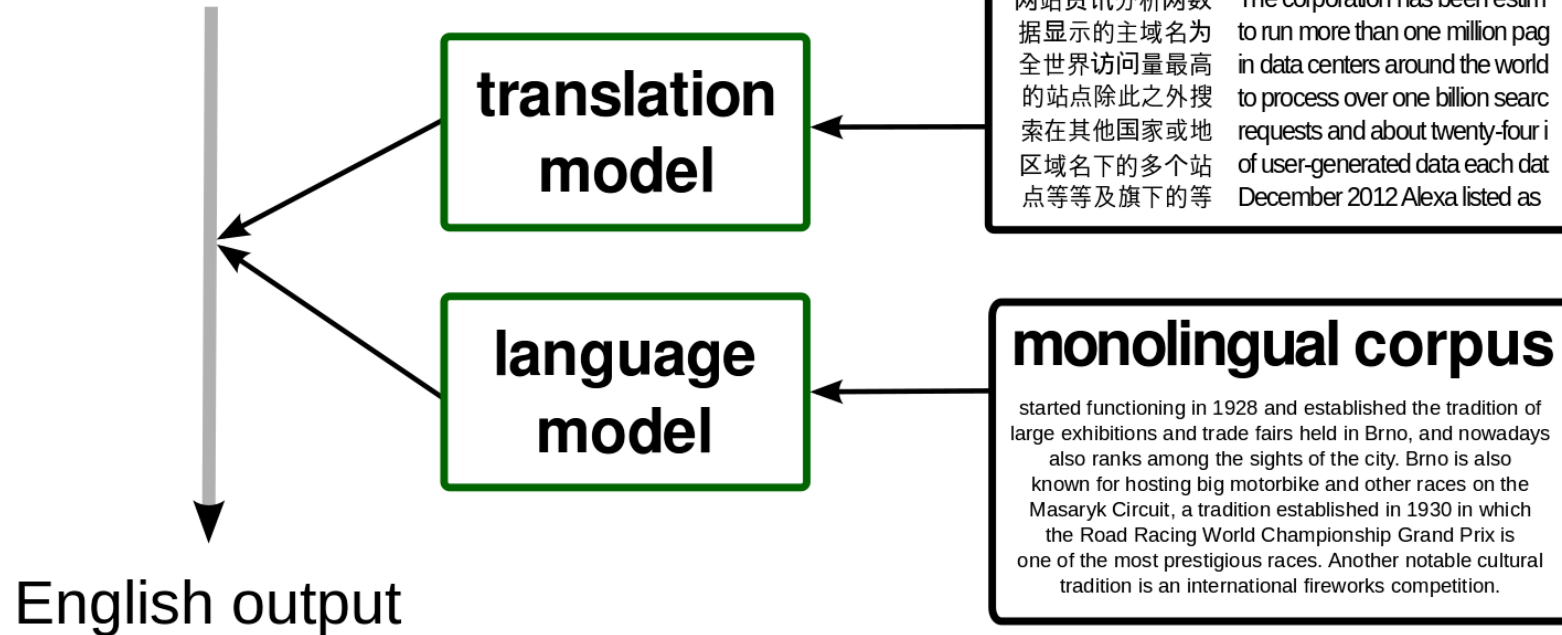
Speech
recognition

Anytime a linguist leaves the group the (speech) recognition rate goes up
- Fred Jelinek

Statistical Machine Translation

- Required a parallel corpora

似乎格式有問題



Rise of statistical learning

- Required human-designed representations and input features

Var	Definition	Value in Fig. 5.2
x_1	count(positive lexicon) $\in$ doc)	3
x_2	count(negative lexicon) $\in$ doc)	2
x_3	$\begin{cases} 1 & \text{if "no"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	log(word count of doc)	$\ln(64) = 4.15$

Can these features be automatically learned?

Promise of deep learning

- Most NLP works in the past focused on human-designed representations and input features

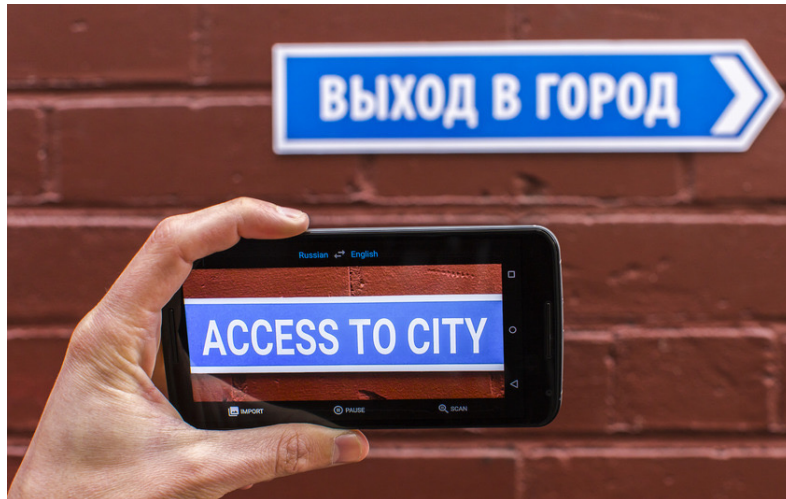
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x_6	log(word count of doc)	$\ln(64) = 4.15$

- **Representation learning attempts to automatically learn good features and representations**
- **Deep learning attempts to learn multiple levels of representation on increasing complexity/abstraction**

Deep learning era

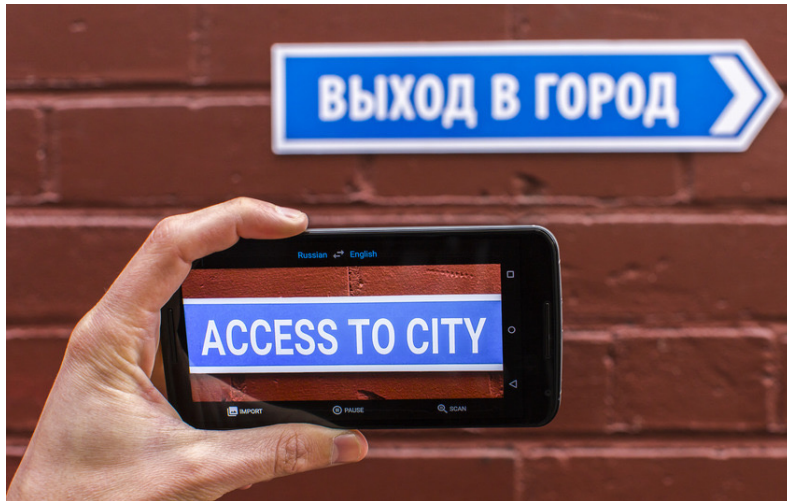
2014-Present: Neural Machine Translation

- Led to significant advances in core NLP technologies



Deep learning era

- Led to significant advances in core NLP technologies
- **Essential ingredient:** large-scale supervision, lots of compute
- Reduced manual effort - less/zero feature engineering



36M sentence pairs

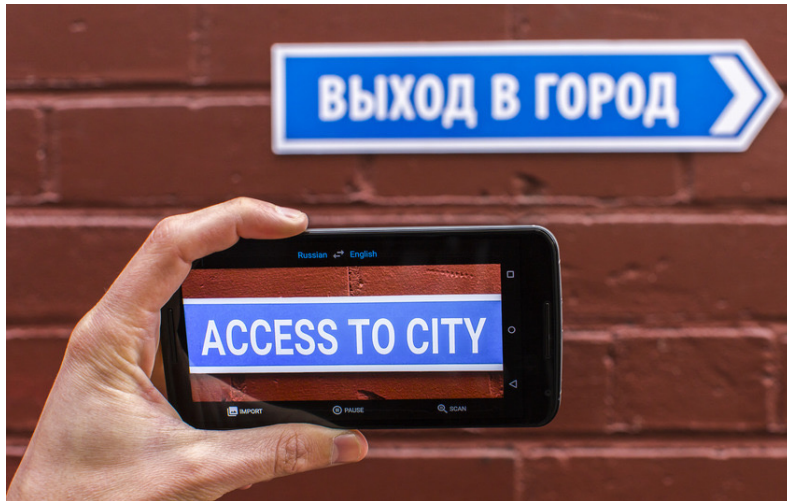
Russian: Машинный перевод - это круто!



English: Machine translation is cool!

Deep learning era

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36M sentence pairs

Russian: Машинный перевод - это круто!



English: Machine translation is cool!

Machine Translation

BBC

Sign in

选项 (英文)

检索

Q

NEWS | 中文

繁

主页

国际

两岸

英国

评论

科技

财经

图辑

音频材料

视频材料

BBC英伦网

巴拿马首任驻华大使专访：与台湾断交之后

他说，不认同中国“买走”台湾邦交国的说法，与中国建立关系对巴拿马有利，不担心影响与美国关系。

🕒 1小时前

巴拿马外交转向周年 中美大国博弈内幕解密

尼加拉瓜运河成谜：人走楼空 “不再提及”

触发萨尔瓦多与台湾断交的港口令美国担忧





人民币贬值、楼价高涨：消费降级中产失去的优质生活

广告



台湾民间发起东京奥运“正名”公投的意义





观点：高铁“一地两检”——中国的强势 港人的无力

江浙：两地价值观和制度差异巨大，冲突无可避免。香港没有反对的权利，也没有反对的能力。



范冰冰消失百日后 中国娱乐业的寒蝉效应

日前发布的《中国影视明星社会责任研究报告》中，最高分徐峥为78分，最低分范冰冰为0分。

🕒 2小时前

中蒙参加俄罗斯军演 “中俄靠拢论”再吸眼球

🕒 2018年 9月 11日

日本人脚踹慰安妇铜像引爆台湾人抗议

🕒 2018年 9月 11日

太空望远镜探测比“三

Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation

Table 10: Mean of side-by-side scores on production data

	PBMT	GNMT	Human	Relative Improvement
English → Spanish	4.885	5.428	5.504	87%
English → French	4.932	5.295	5.496	64%
English → Chinese	4.035	4.594	4.987	58%
Spanish → English	4.872	5.187	5.372	63%
French → English	5.046	5.343	5.404	83%
Chinese → English	3.694	4.263	4.636	60%

(Wu et al., 2016)

Machine Translation

 [Sign in](#) Option (English)

NEWS | 中文 Traditional

[Homepage](#) [International](#) [Cross-strait](#) [United Kingdom](#) [comment](#) [Technology](#) [Finance](#) [Picture series](#) [Audio material](#)

Interview with Panama's first ambassador to China: After breaking diplomatic relations with Taiwan

He said that he does not agree with China's saying that "buy" Taiwan's diplomatic relations with China. Establishing relations with China is beneficial to Panama and does not worry about affecting relations with the United States.

🕒 1 hour ago

Panamanian Diplomacy Turns to Anniversary

The Nicaragua Canal is a mystery: people go to the floor and "no longer mention"

The port that triggered the break of El Salvador and Taiwan has worried the United States



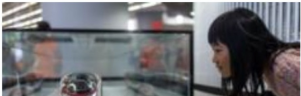


Renminbi depreciation, property prices are rising: consumption downgrades the loss of quality life in the middle class

ADVERTISING

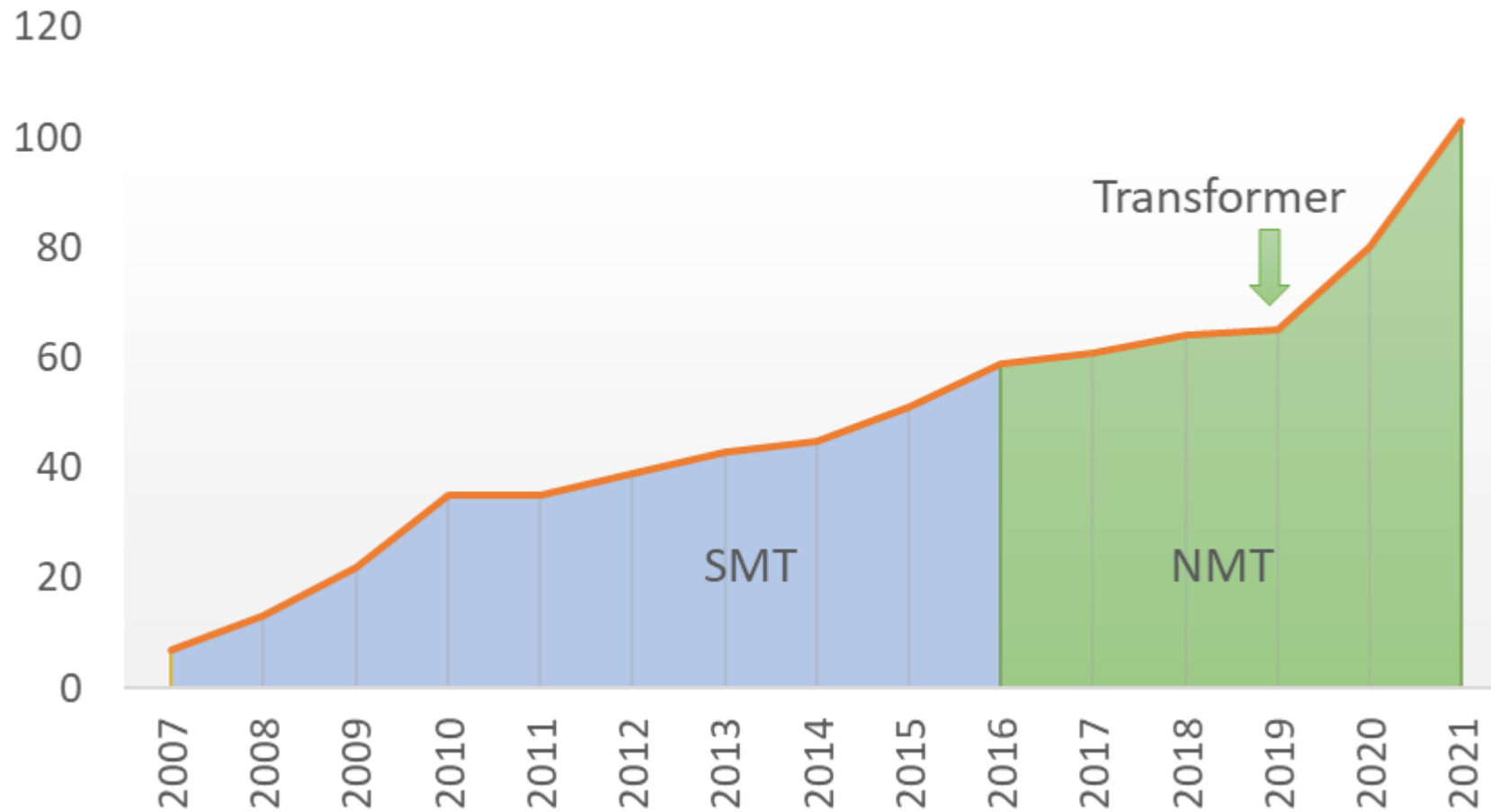


The significance of the Taiwanese people's "referred to" referendum



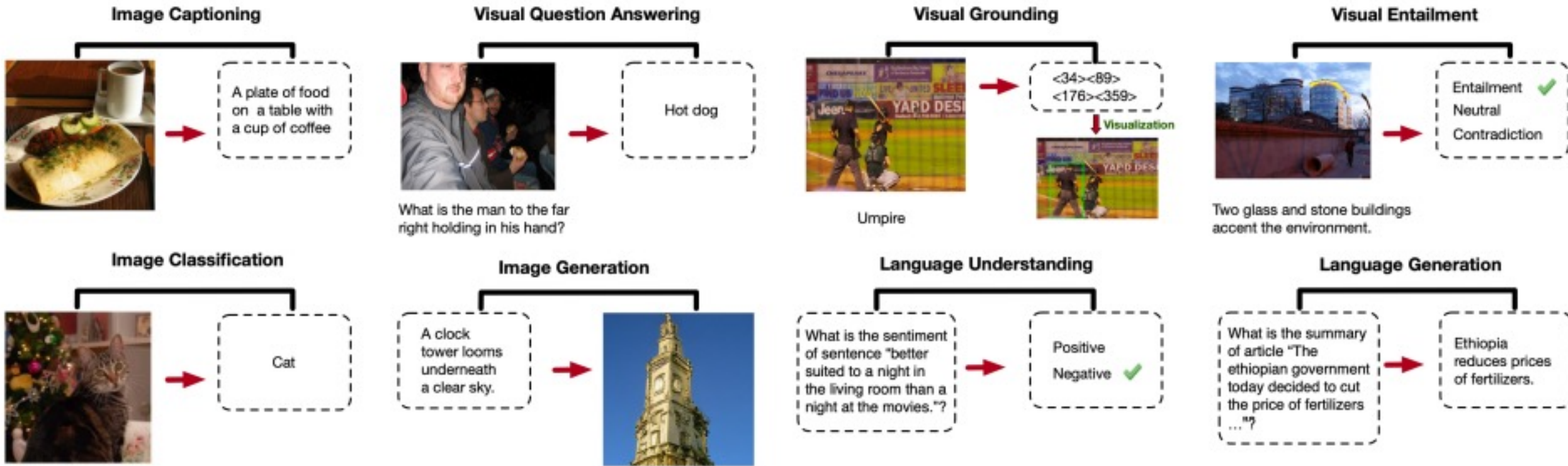
China and Mongolia participate in the Russian military

Machine Translation (across 100 languages)



<https://www.microsoft.com/en-us/research/blog/microsoft-translator-now-translating-100-languages-and-counting>

Multimodal models



<https://github.com/OFA-Sys/OFA>

OFA (Wang et al, 2022)

Power of Data

CleverBot (2010)

User:	Do you like Doctor Who?
Cleverbot:	Yes, so funny.
User:	What's your favorite show?
Cleverbot:	I don't have one.
User:	What's your name?
Cleverbot:	Jason I already told you.
User:	No your real name.
Cleverbot:	Stephanie IS my real name.
User:	I thought it was Jason?
Cleverbot:	How did you come to that conclusion.
User:	Because you said so.
Cleverbot:	I have to do my homework.

<https://www.cleverbot.com/>

How it works:

- Corpus of conversational turns
- Find the most similar sentence and copy the response
- Learn from human input

What do you get?

- Something that someone say
- Incoherent conversation

Power of Data

Meena (Google, 2020)

Conversation D

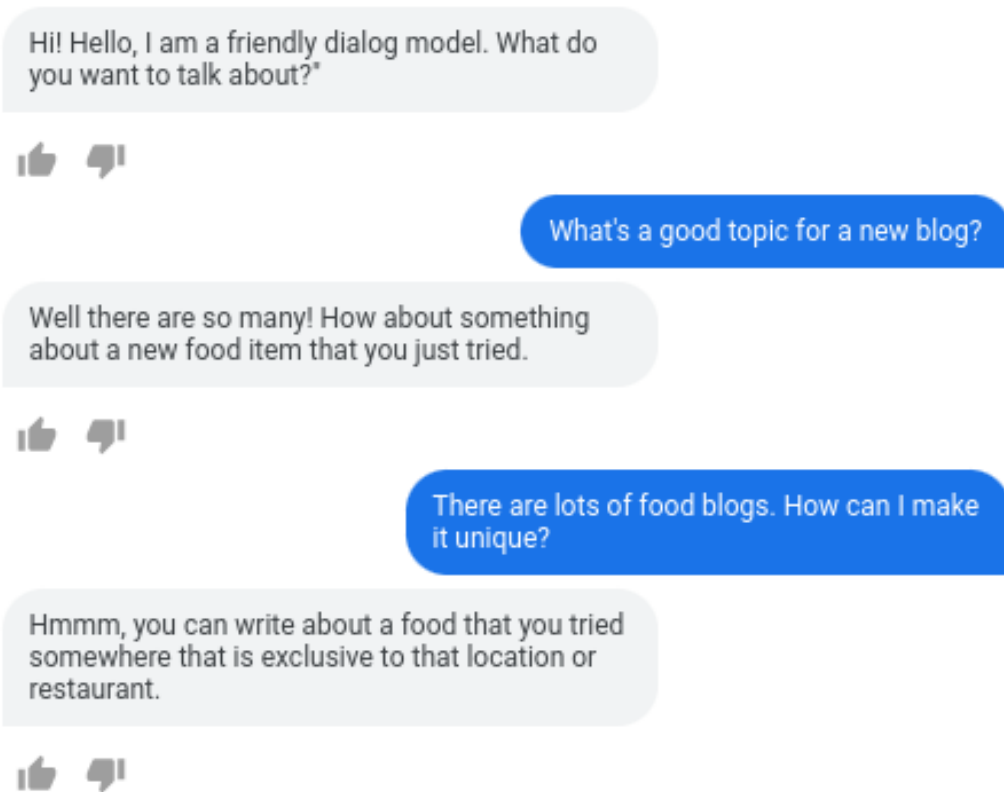
Human: *I like how cute cows are*
Meena: *Ah, interesting. You like how they are cute, right?*
Human: *yeah, and they're really smart*
Meena: *I heard they go to college*
Human: *cows go to college?*
Meena: *I heard that a cow went to Harvard.*
Human: *what did the cow study?*
Meena: *Bovine sciences.*
Human: *do horses go to Harvard?*
Meena: *Horses go to Hayvard.*
Human: *that's a pretty good joke, I feel like you led me into it*
Meena: *You were trying to steer it elsewhere, I can see it.*

How it works:

- Corpus of conversational turns (over 40B words)
- Train huge neural network (2.6 billion parameters) for 30 days on 2048 TPUs cores
- Predict response given a sentence

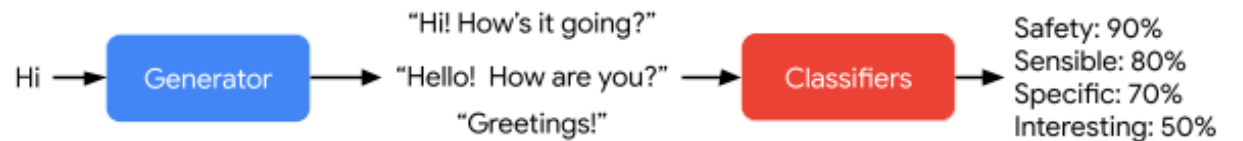
Power of Data

LaMDA (Google, 2022)



How it works:

- Pretrain on even more data
- Fine-tune so it has certain characteristics



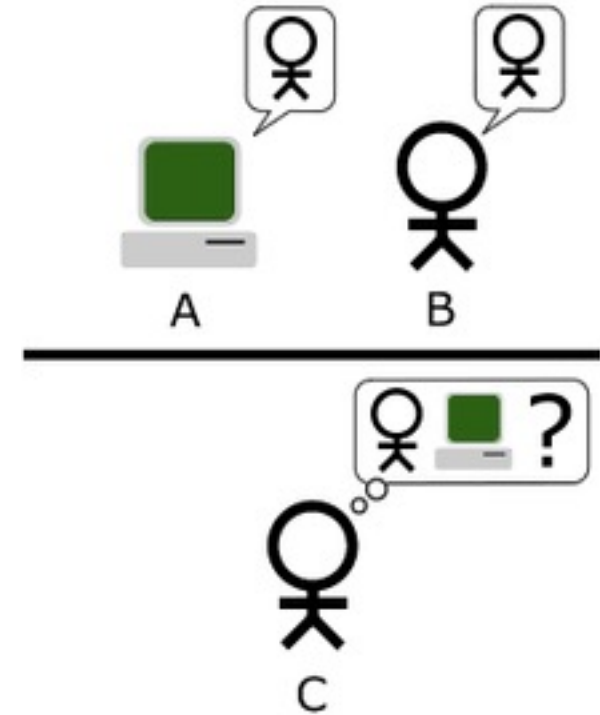
Is this chatbot sentient?

<https://ai.googleblog.com/2022/01/lamda-towards-safe-grounded-and-high.html>

https://youtu.be/CvD9U_3hBPo

Turing Test

Imagine an "**Imitation Game**," in which a man and a woman go into separate rooms and guests try to tell them apart by writing a series of questions and reading the typewritten answers sent back. In this game both the man and the woman aim to convince the guests that they are the other.



Can you guess:
Computer or human?



Alan Turing

We now ask the question, "**What will happen when a machine takes the part of A in this game?**" Will the interrogator decide wrongly as often when the game is played like this as he does when the game is played between a man and a woman? These questions replace our original, "**Can machines think?**"

Turing test solved?

Talking to Google Duplex: Google's human-like phone AI feels revolutionary

Believe the hype—Google's phone-call bot is every bit as impressive as promised.

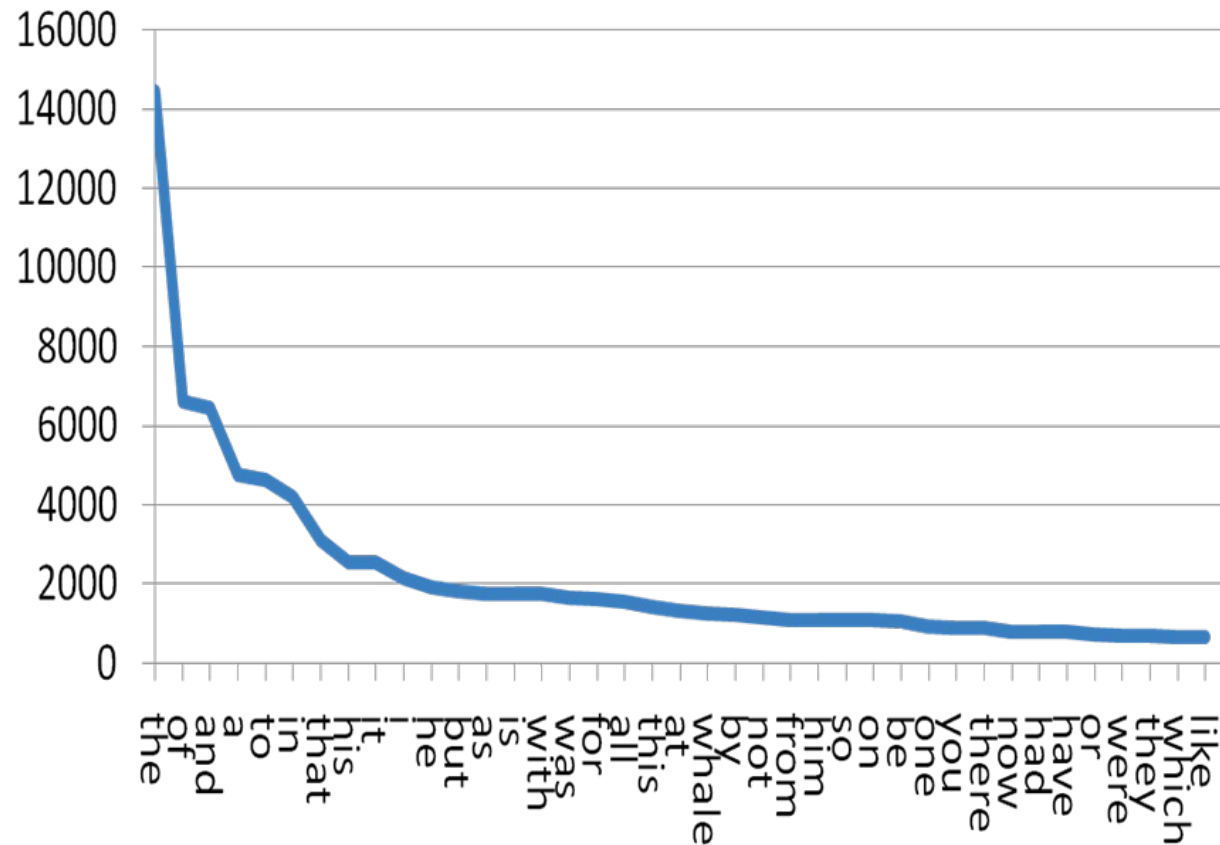


<https://www.youtube.com/watch?v=D5VN56jQMWM&feature=youtu.be&t=70>

Many remaining challenges in NLP!

Challenges in modern NLP

- *Scale*: Large number of phenomena
- *Sparsity*: Text data is often heavy-tailed



Challenges in modern NLP

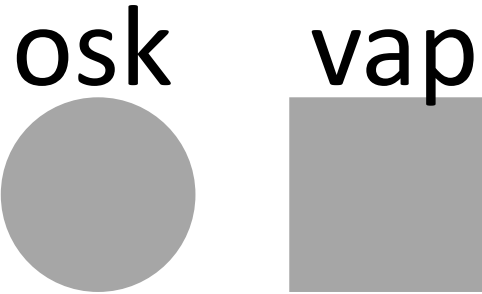
- *Bias*: Models learn biases in available data



- *Context:* Knowledge bases, perception, interaction

Compositional generalization

Grounding



Compositionality



Generalization

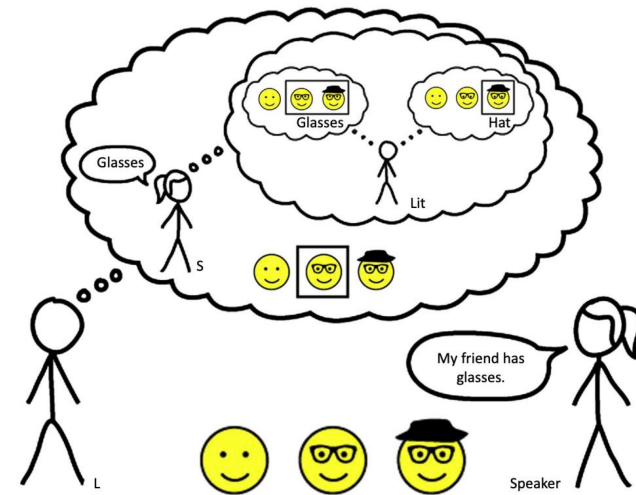


Challenges of understanding language

- Does solving language require solving AI?
- Language derives its meaning from use



'block' 'pillar' 'slab' 'beam'.



- To truly understand language: need **context**, **world knowledge**, **mental model** of other people



ALFRED

A Benchmark for Interpreting
Grounded Instructions for Everyday Tasks

<https://askforalfred.com/>

What does it mean to understand language?
A look at levels of linguistic representations

Translation

Dream of universal translator

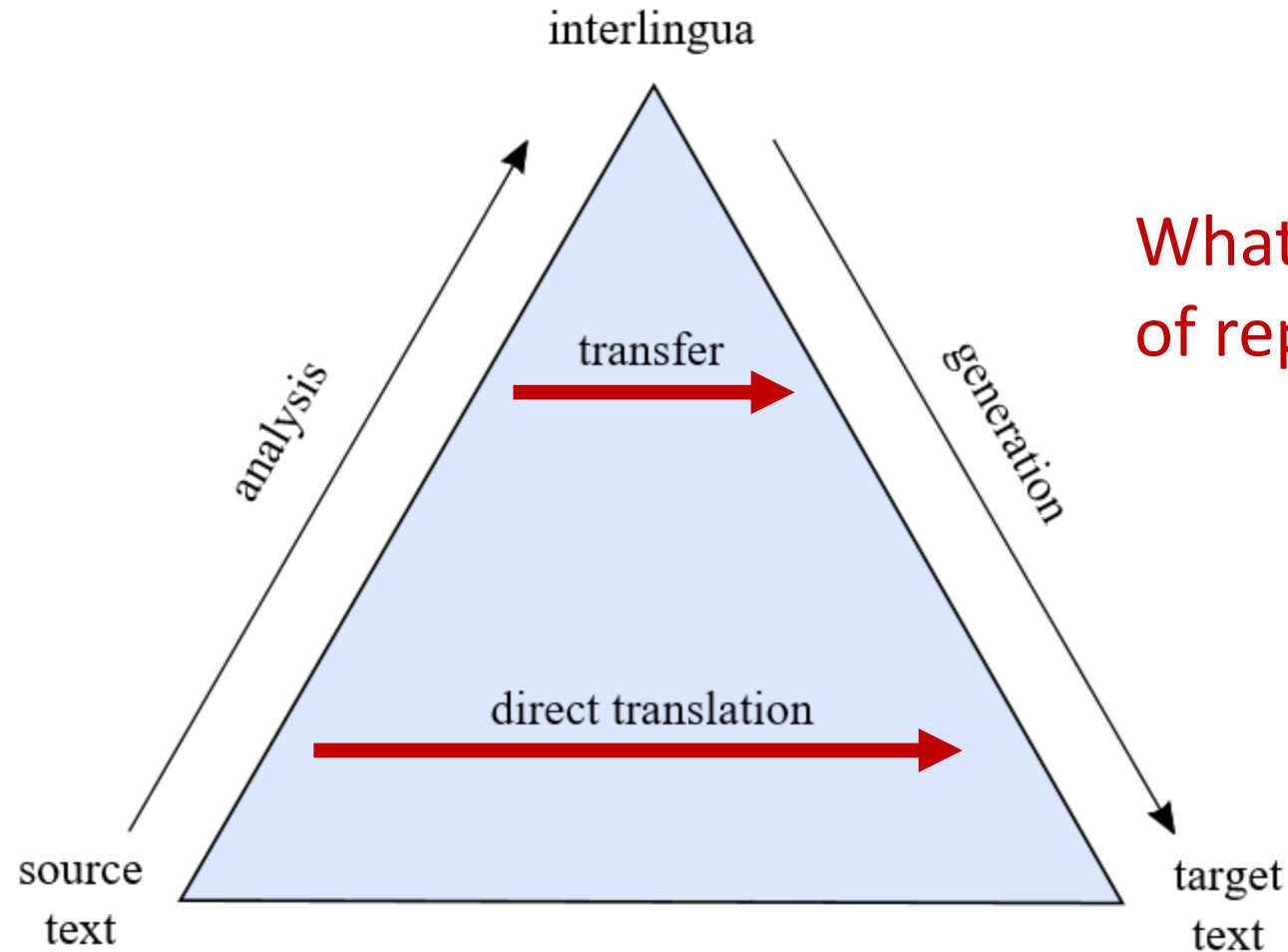
- Different languages: Machine translation

the cat sat on the table  η γάτα κάθισε στο τραπέζι

Not just between languages, but across modalities as well

- Audio to text: Speech recognition
- Image to language: Image captioning

Vauquois Triangle for translation

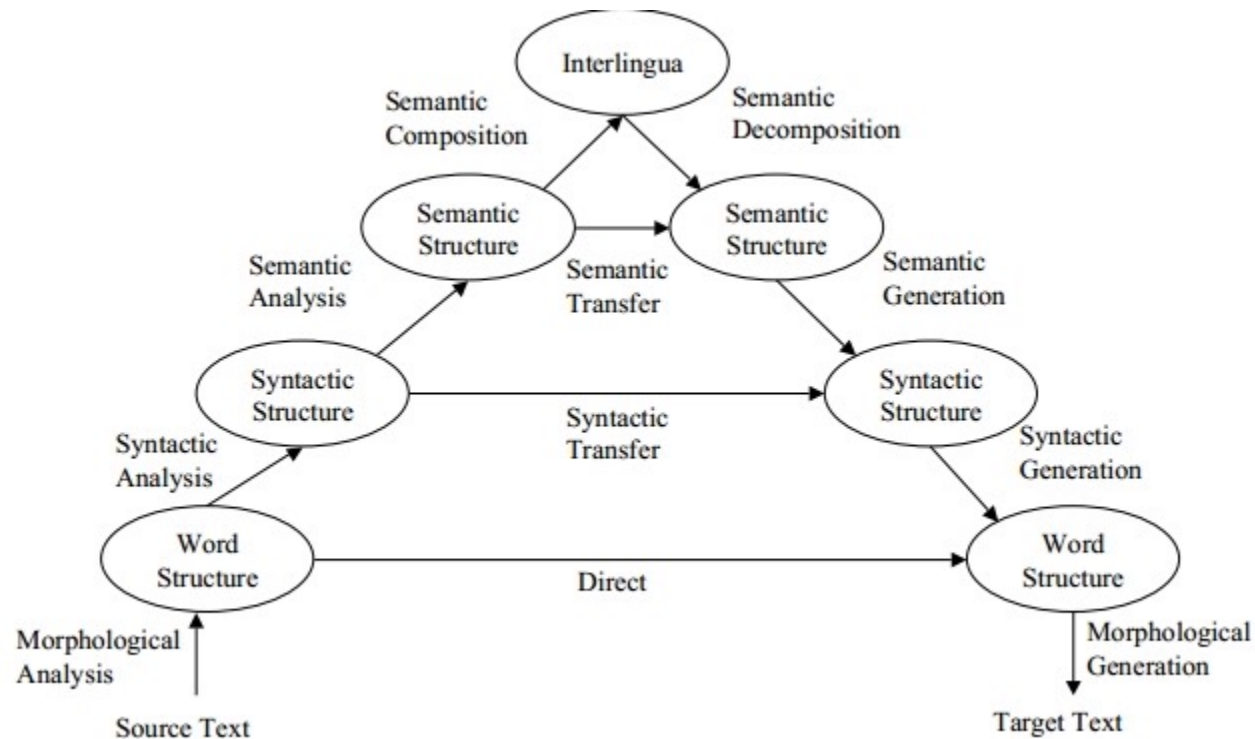


What are the levels of representation?

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Vauquois Triangle for translation



Pragmatics

Semantics

Syntax

Morphology

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Levels of language

- Phonetics: acoustic and perceptual elements
- Phonology: basic sounds (phonemes) and rules for combination
 - e.g. syllable constraints in Japanese.
 - e.g. English loanword for “dog” in Japanese is “doggu”
- Morphology: how morphemes combine to form words, relationship of phonemes to meaning
 - e.g. delight-ed vs. de-light-ed

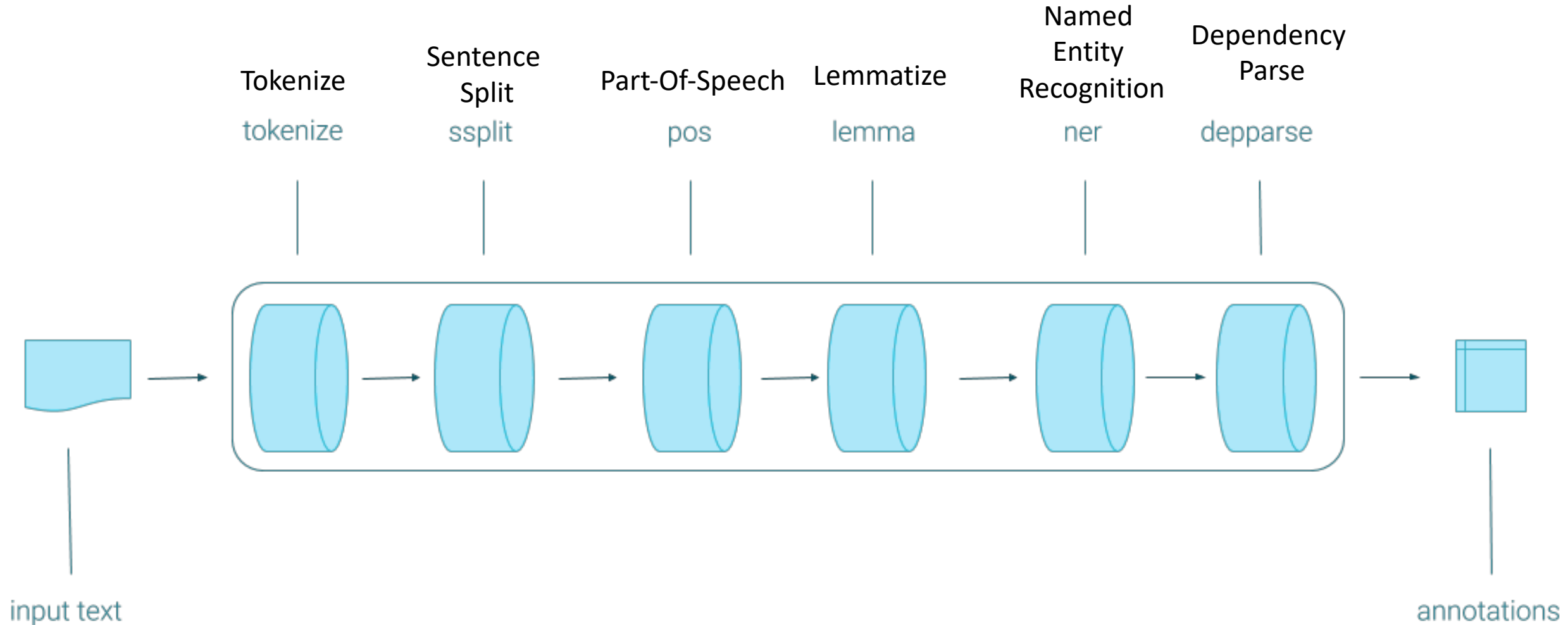
More details in option video lecture (to be posted on Canvas)

Levels of language

- Syntax: set of rules, principles, and processes that govern the structure of sentences in a given language
 - sentence formation
 - o e.g. The clown who the musician hits watches the ballerina
- Semantics: focuses on the relation between *signifiers*, like words, phrases, signs, and symbols, and what they stand for; their denotation
 - meaning (from syntax to logical formulas)
 - e.g. Everyone is not here => what does this mean? Nobody / Not everyone is here.
- Pragmatics: studies the ways in which context contributes to meaning
 - e.g. “The ham sandwich wants another coffee”

More details in option video lecture (to be posted on Canvas)

Traditional NLP Pipeline



More details in option video lecture (to be posted on Canvas)

Annie, the kitten was born at night. She was a gray furball.

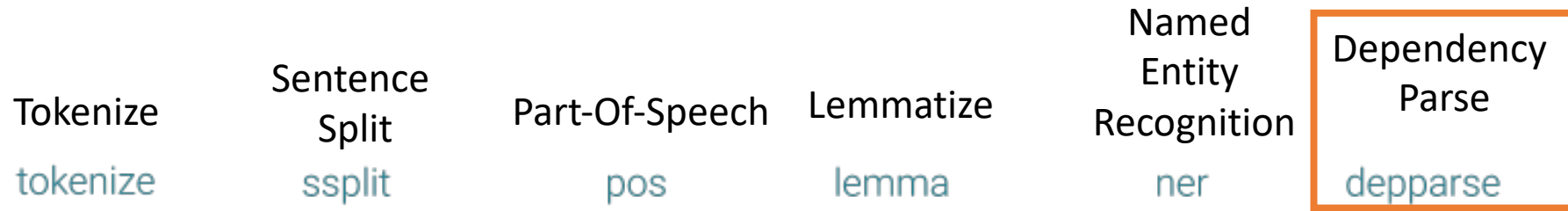
Traditional NLP Pipeline



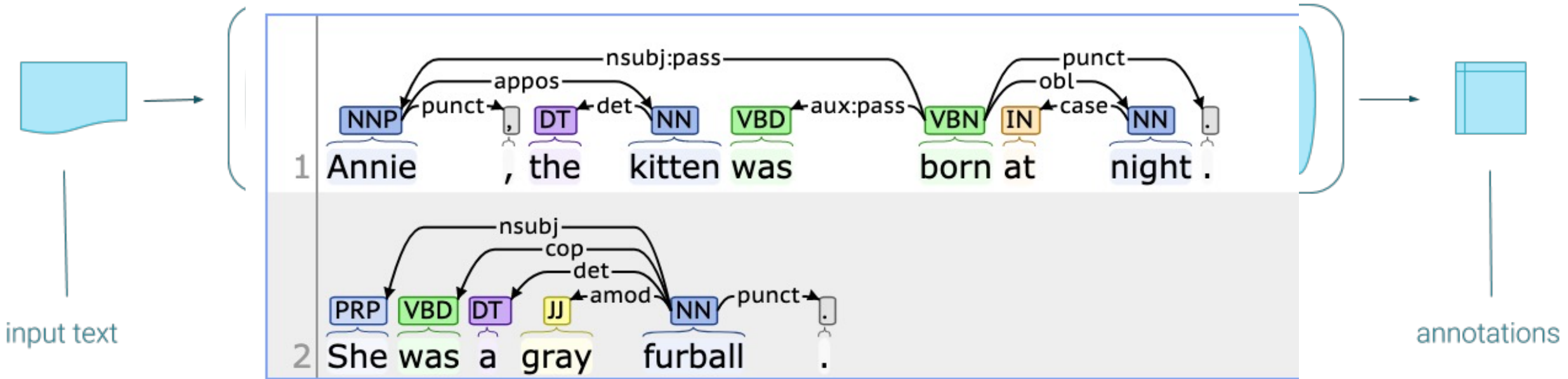
More details in option video lecture (to be posted on Canvas)

Annie, the kitten was born at night. She was a gray furball.

Traditional NLP Pipeline



Basic Dependencies:



More details in option video lecture (to be posted on Canvas)

Annie, the kitten was born at night. She was a gray furball.

Try out some NLP systems for yourself!

- Spacy (<https://spacy.io/>)
 - Interoperates with Stanza and AllenNLP
- Stanford CoreNLP / Stanza
 - CoreNLP (<http://stanfordnlp.github.io/CoreNLP/>)
 - Stanza (<https://stanfordnlp.github.io/stanza>)
- AllenNLP (<https://allennlp.org/>)

What will this course cover?

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Representations

- Embeddings
- Compositional
- Structured

Methods

- Rule based
- Statistical Methods
- Neural Models
- Dynamic programming
- Language Models

Applications and Tasks

- Machine Translation
- Dialogue Agents
- Information Extraction
- Question Answering
- Grounding

What will this course cover?

Representations

- Embeddings
- Compositional
- Structured



early November

Methods

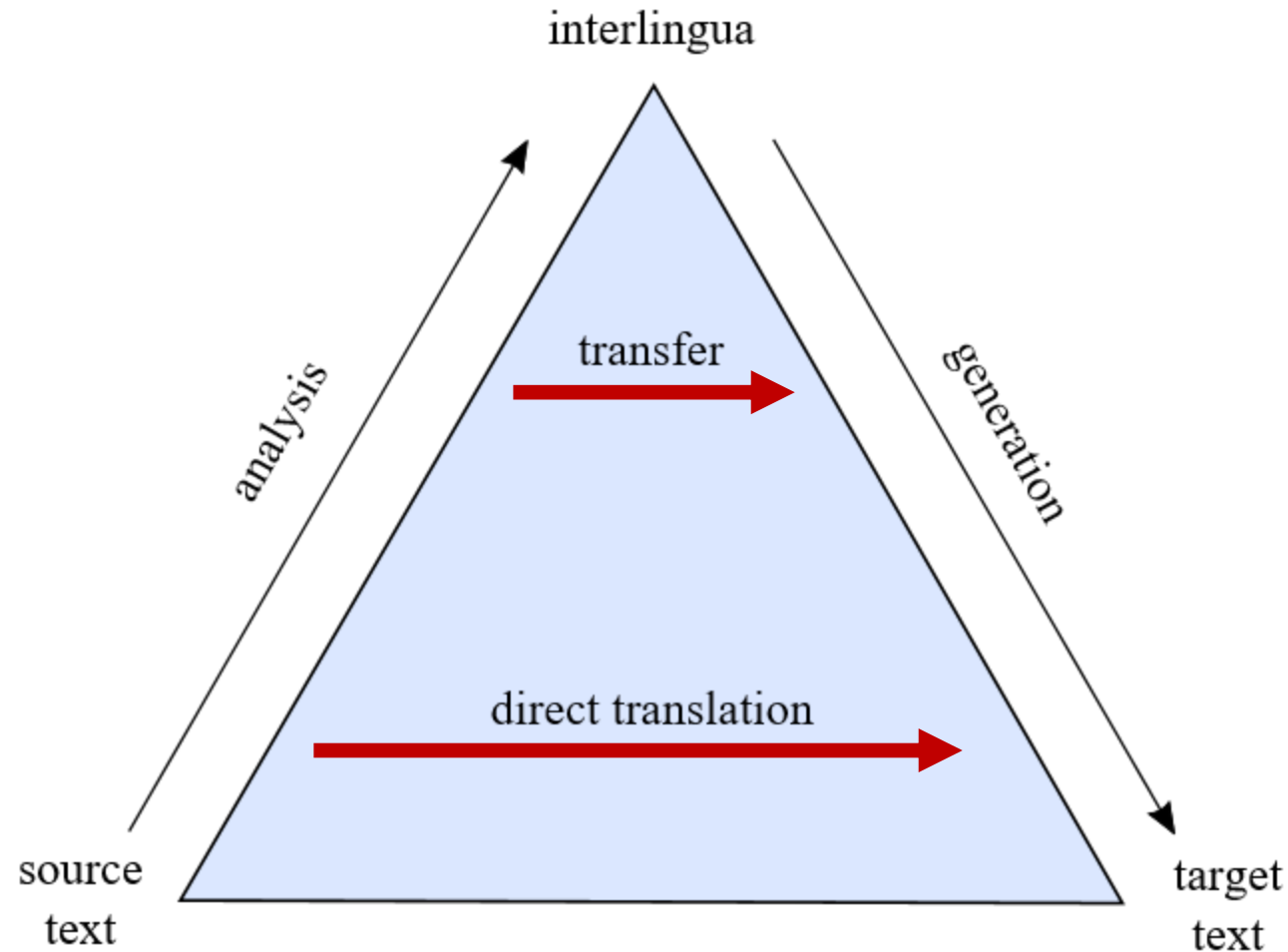
- Rule based
- Statistical Methods
- Neural Models
- Dynamic programming
- Language Models

Applications and Tasks

- Machine Translation
- Dialogue Agents
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later part of the course
(mid November)

Vauquois Triangle for translation



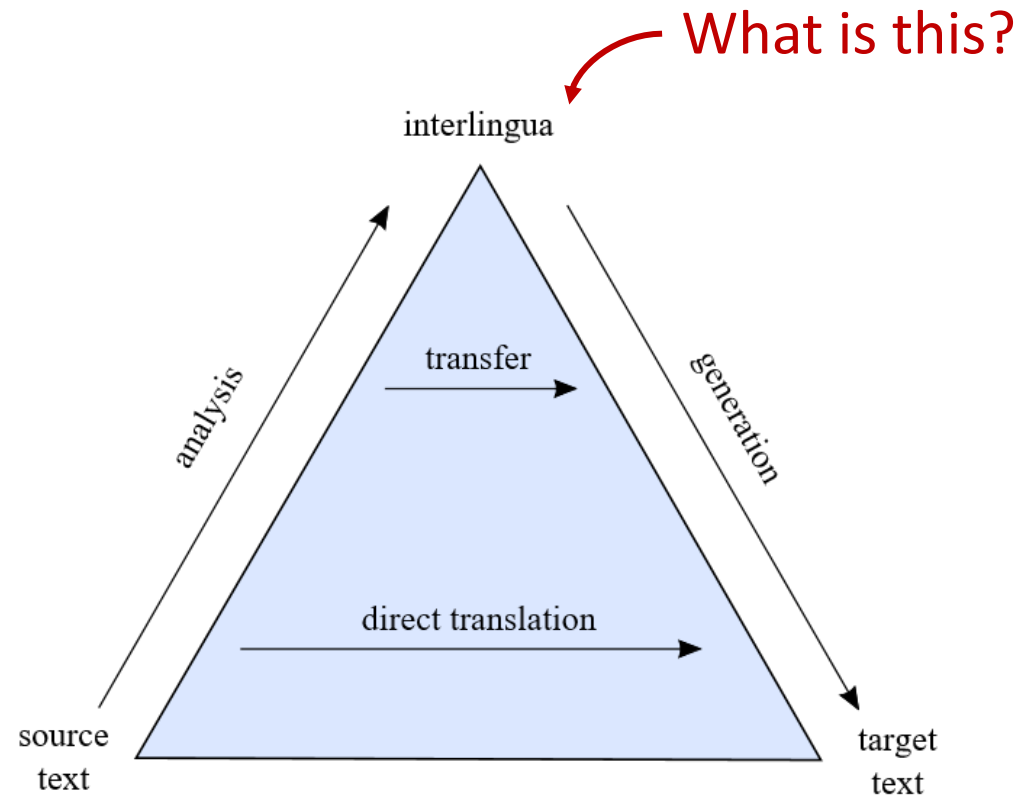
the cat sat on the table

η γάτα κάθισε στο τραπέζι

Key Questions

- How to **represent** language?
- How to **compose** meaning?
- How do we translate from the “surface form” of language to a representation that a computer can process, and back?
 - How do we **encode** / **parse** the meaning of a sentence?
 - How do we **decode** / **generate** text from a meaning representation?
- What are the key methods / algorithms for **learning** these representations?

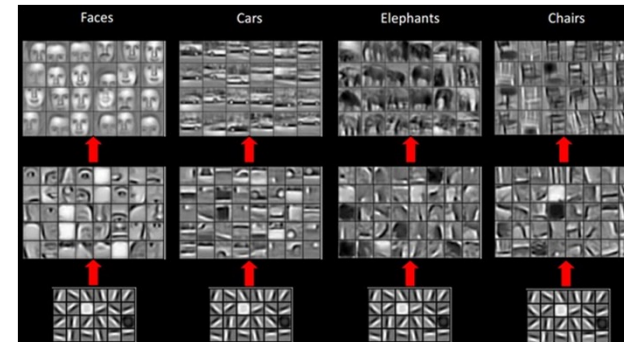
What should the representation look like?



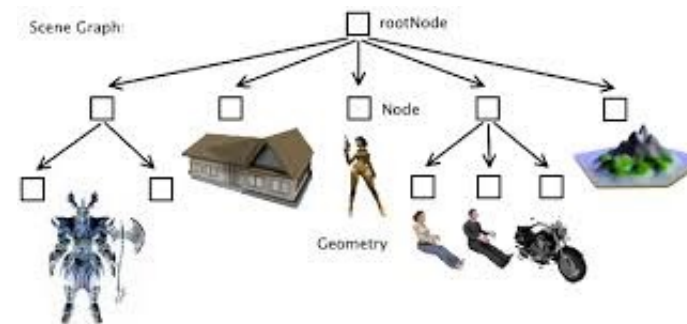
$$L = \begin{bmatrix} \vdots & \vdots & \vdots & \dots & \vdots & \vdots \end{bmatrix}_n$$

the cat mat ...

$|V|$



Lambda expression:
 $((\lambda x . (\lambda y . x)) (\lambda z . z)) (\lambda a . a)$



Representations

How to represent the meaning of something?



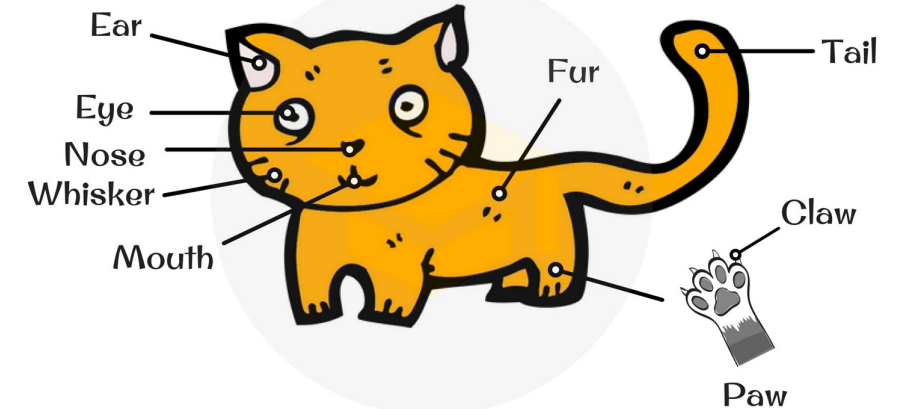
“cat”

cat: a small domesticated carnivore, *Felis domestica* or *F. catus*, bred in a number of varieties.

```
cat → {  
  isMammal: true  
  hasFur: true  
  hasLegs: true  
  meows: true  
  barks: false  
  height: 9.1 – 9.8 in  
  weight: 7.9 – 9.9 lbs  
  ...  
}
```

Attributed
representation

Parts of a cat



TESL.COM

Representations



“cat”



“dog”

Representing meaning as vectors

- common representation space
- enables information sharing
- can be learned from data

- One-hot

cat = [0 0 0 1 0 0 0]

dog = [0 0 0 0 0 1 0]

- Embeddings

cat = [0.04 1.79 -1.79 1.07 0.48]

dog = [0.61 1.84 -1.12 0.52 0.53]

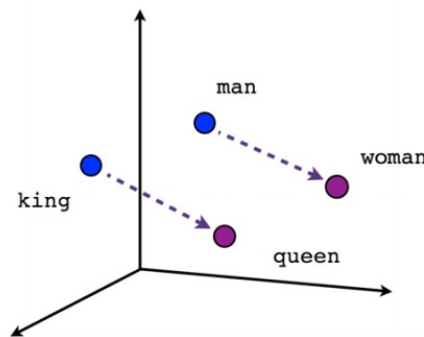
Distributed representations

Project words onto a
continuous vector space

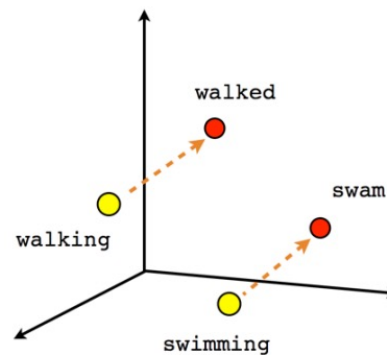
Similar words closer to each other



kitten
cat
dog



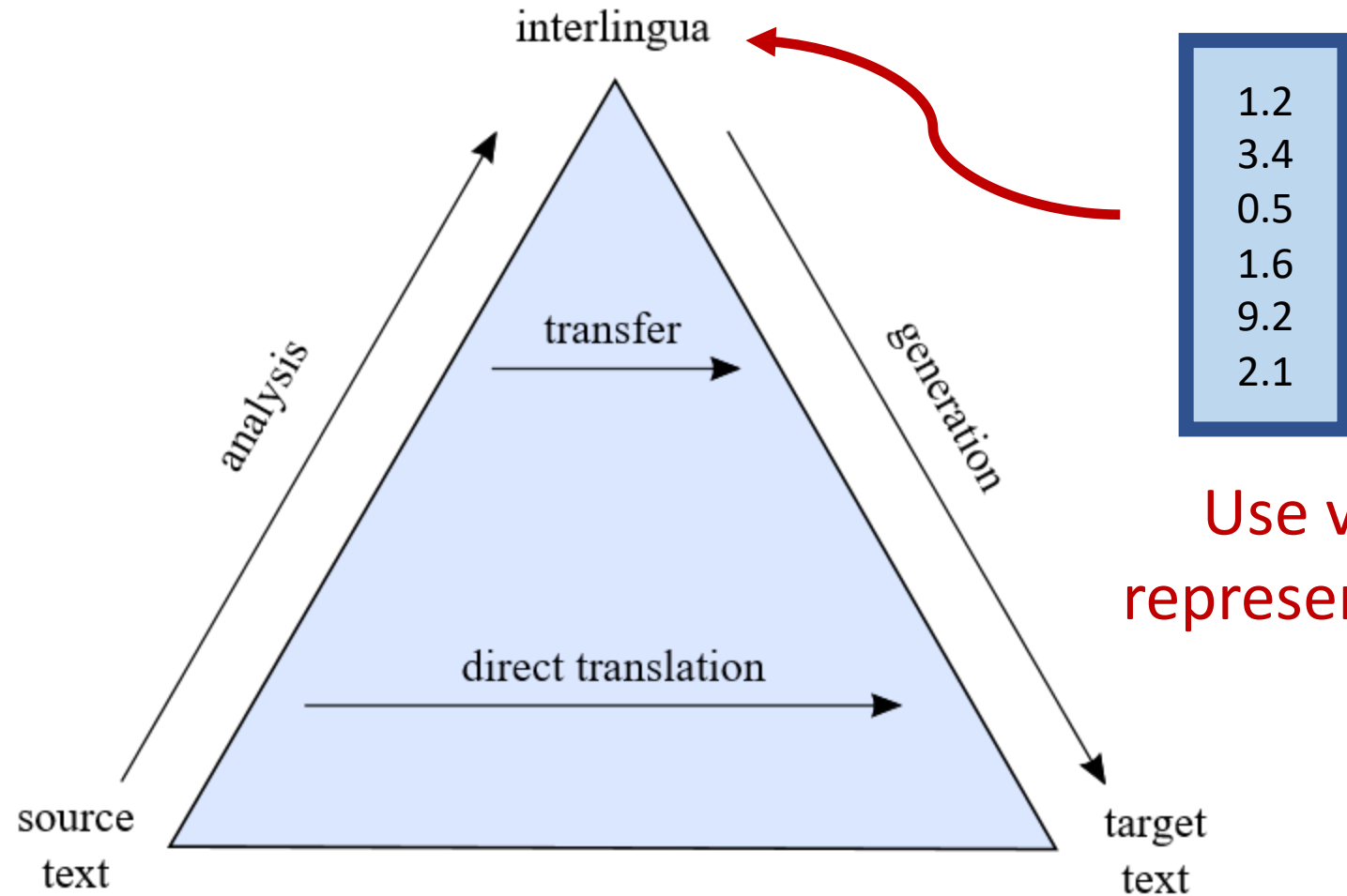
Male-Female



Verb tense

$$v(\text{king}) - v(\text{man}) + v(\text{woman}) = v(\text{queen})$$

Vauquois Triangle for translation

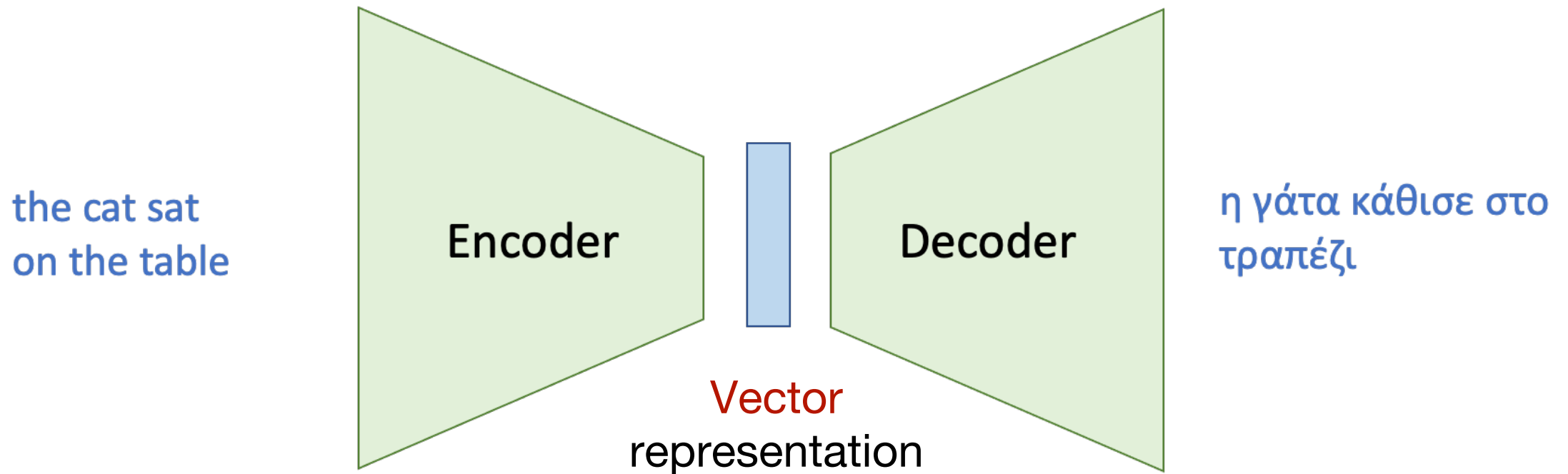


Use vector to
represent meaning

the cat sat on the table

η γάτα κάθισε στο τραπέζι

Vector based representation



Common choice: flat unstructured vector of real numbers

- easy to work with
- well defined mathematical operations
- can be used to measure if two things are similar or not

Translating across modalities

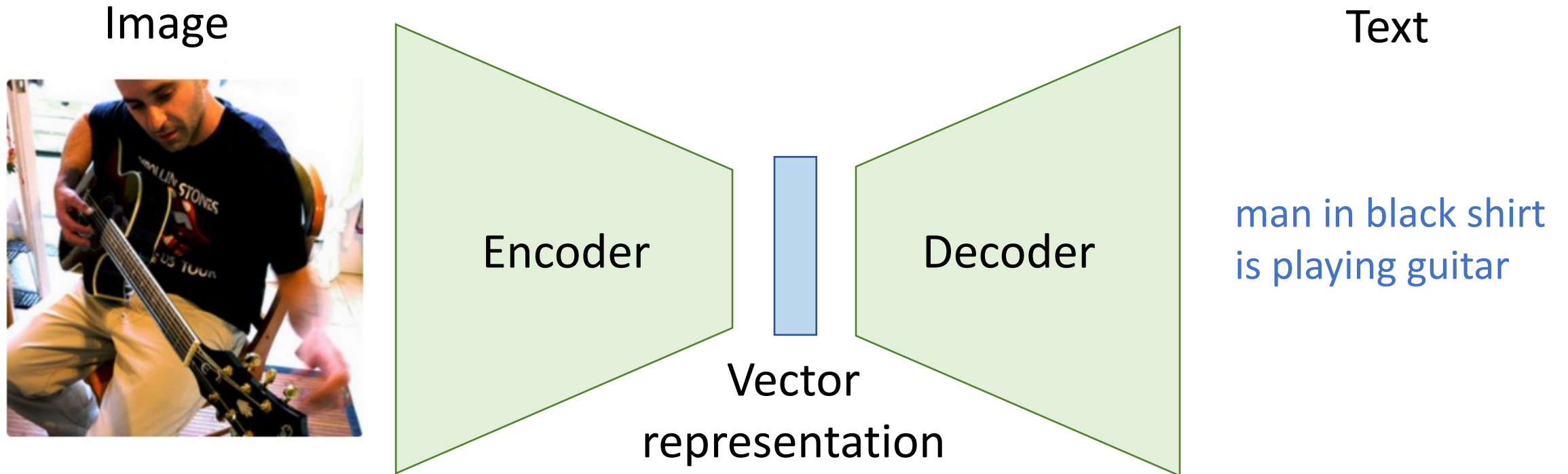
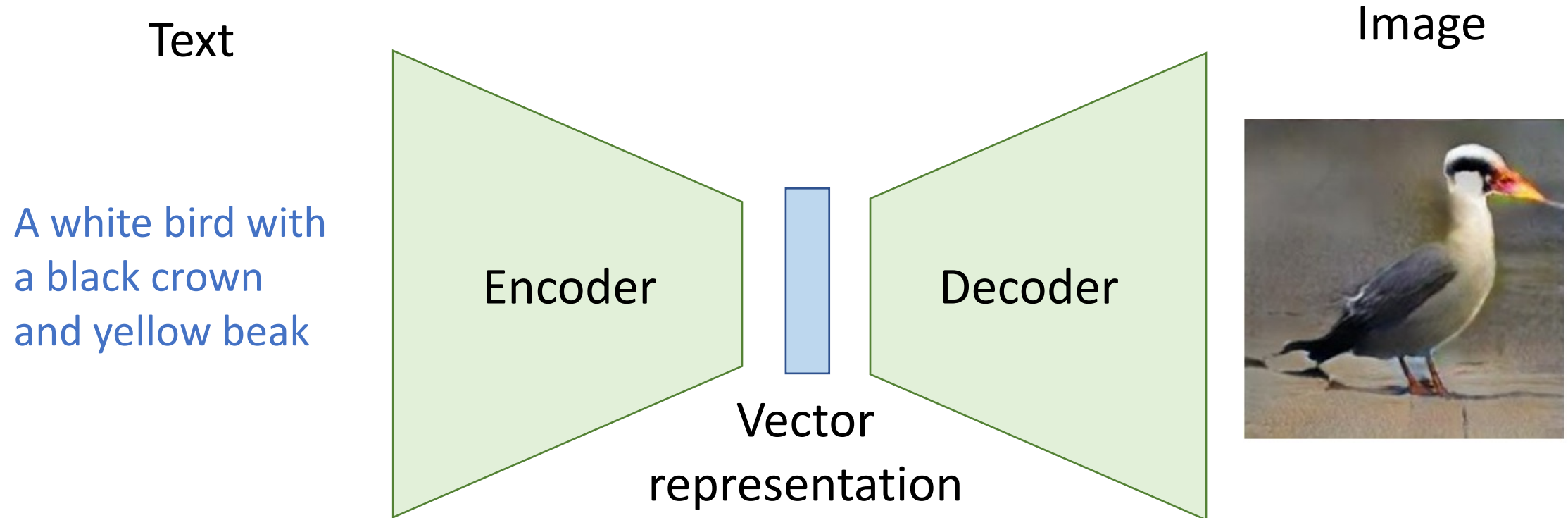


Image captioning

“Deep Visual-Semantic Alignments for Generating Image Descriptions”
[Karpathy and Fei-Fei CVPR 2015]

Translating across modalities



“StackGAN: Text to Photo-realistic Image Synthesis with Stacked Generative Adversarial Networks”
[Zhang et al, ICCV 2017]

Translating across modalities

Text

Brown colored dining table.
It has four legs made of wood.

Encoder

Vector
representation

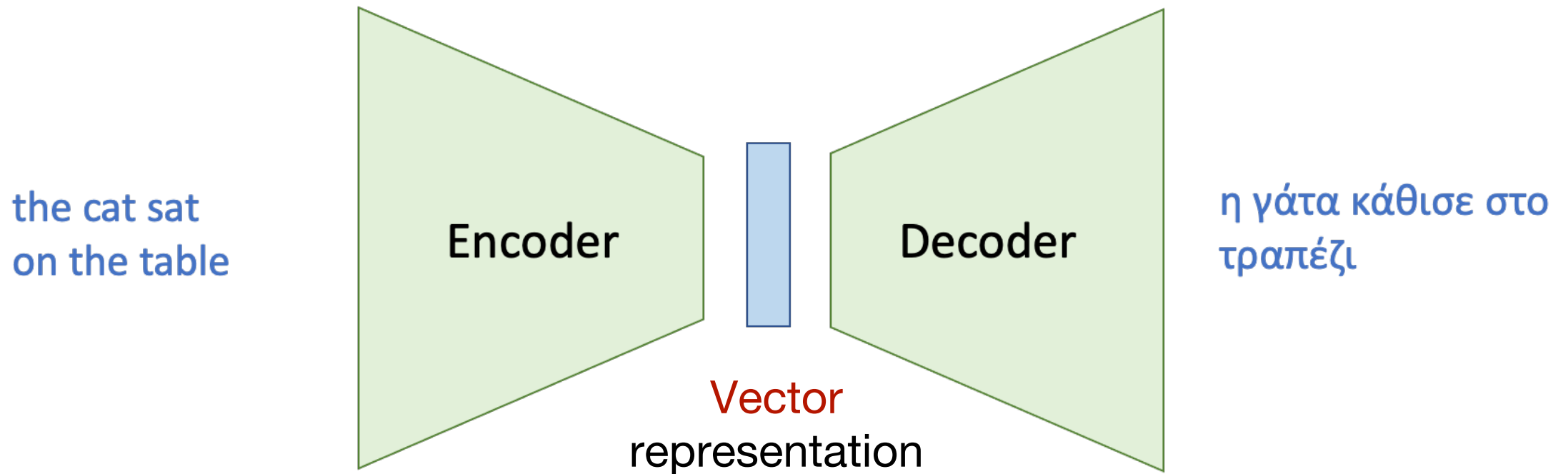
Decoder

3D Shape



“Text2Shape: Generating Shapes from Natural Language by Learning Joint Embeddings”
[Chen et al, ACCV 2018]

Vector based representation



Common choice: flat unstructured vector of real numbers

- easy to work with
- well defined mathematical operations
- can be used to measure if two things are similar or not

How to **encode** language?

What is this function that takes the sentence and converts it into a vector?

the cat sat
on the table

Encoder

Vector
representation

Encoder

- How to convert the sentence into a meaning representation?

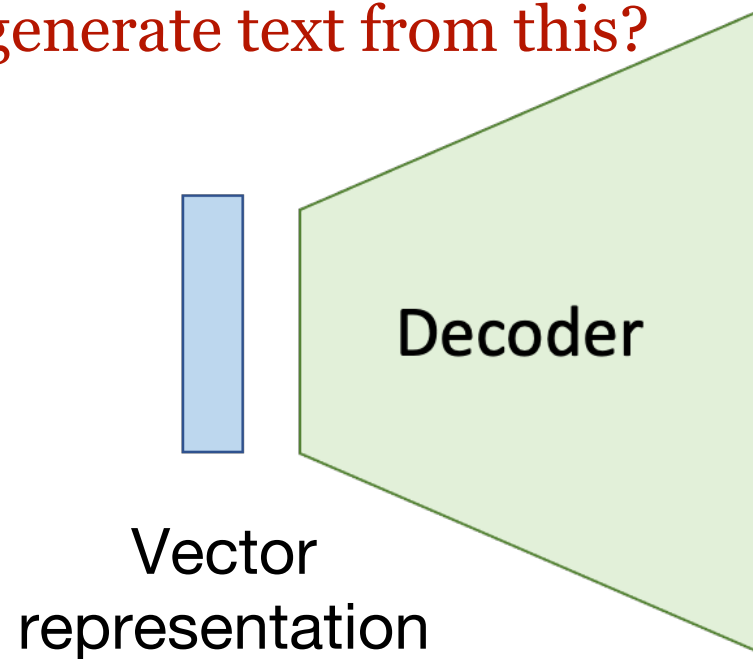
Understanding what is said
(encoding, parsing, feature extraction)

How to **decode** language?

How do we generate text from this?

Decoder

- How to take the vector representation and generate a sequence of words?

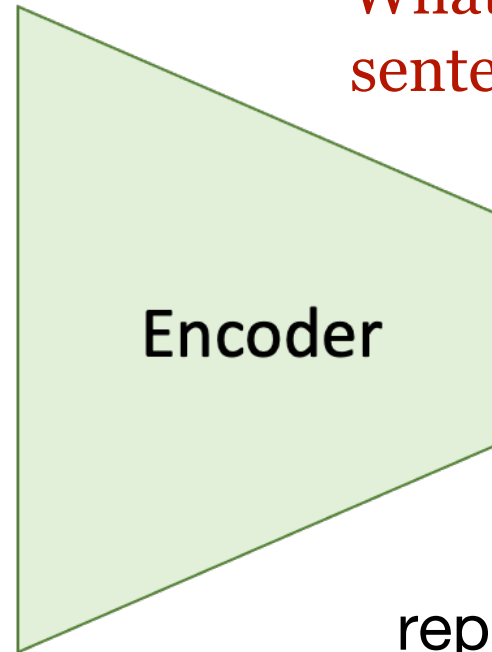


Deciding what to say
(decoding, generating)

How to **encode** language?

What is this function that takes the sentence and converts it into a vector?

the cat sat
on the table



Vector
representation

Properties we want

- Be able to represent a single word
- Be able to represent the meaning of a sentence as a composition of the meaning of each word

Understanding what is said
(encoding, parsing, feature extraction)

How to learn representations for **words**?



start with representing words as discrete tokens

- One-hot

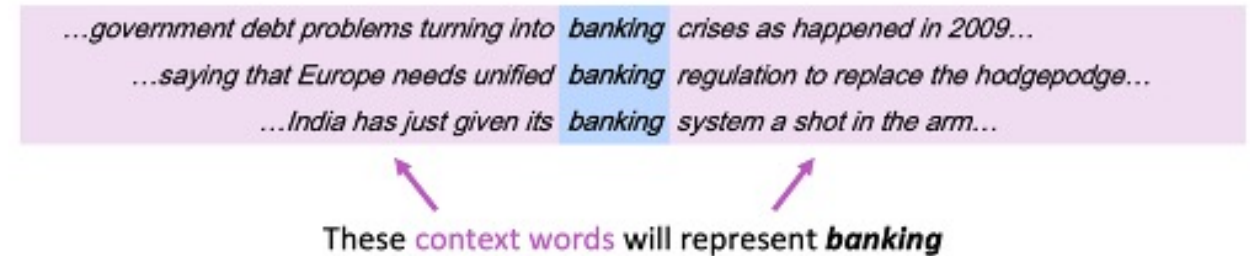
cat = [0 0 0 1 0 0 0]
dog = [0 0 0 0 0 1 0]

learn about neural models that use vectors

- Embeddings

cat = [0.04 1.79 -1.79 1.07 0.48]
dog = [0.61 1.84 -1.12 0.52 0.53]

use **context** to represent words



learn a dense vector representation that can be used to **predict** words from their context

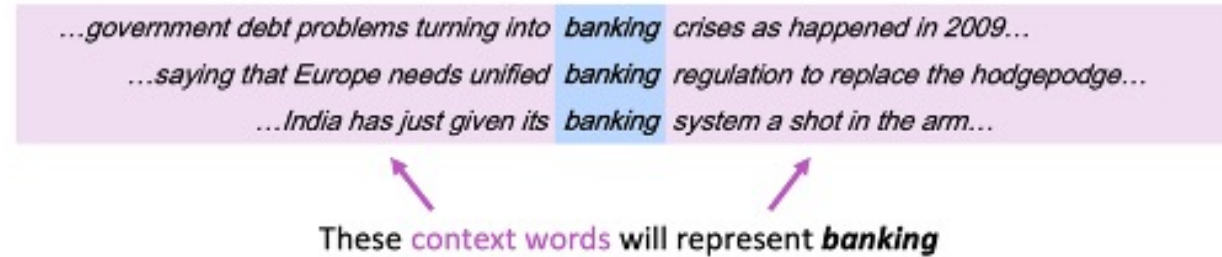
- Predict what goes next

The cat sat on the _____

- Fill in the blank

A bottle of _____ is on the table.

How to learn representations



- Pretrain on large amount of text
- During pretraining: learn to predict words from their context

- Predict what goes next

The cat sat on the _____

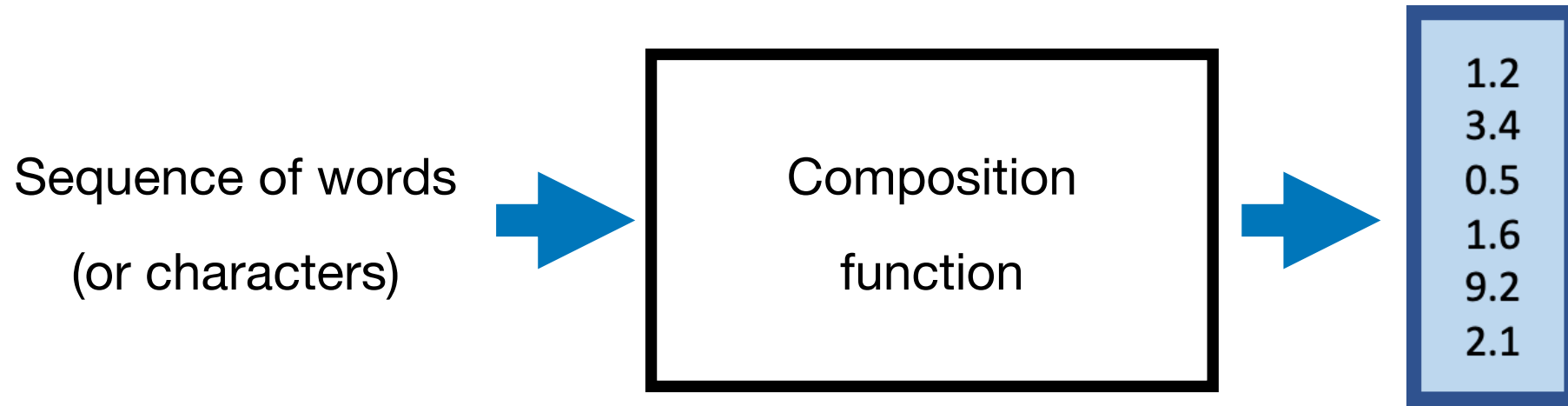
Language Model $P(w_t | w_{<t})$

- Fill in the blank

A bottle of _____ is on the table.

- Use pretrained model and fine-tune it for downstream tasks

How to **compose** language?



Bag of Words (BOW)

Window BoW (bag of ngrams)

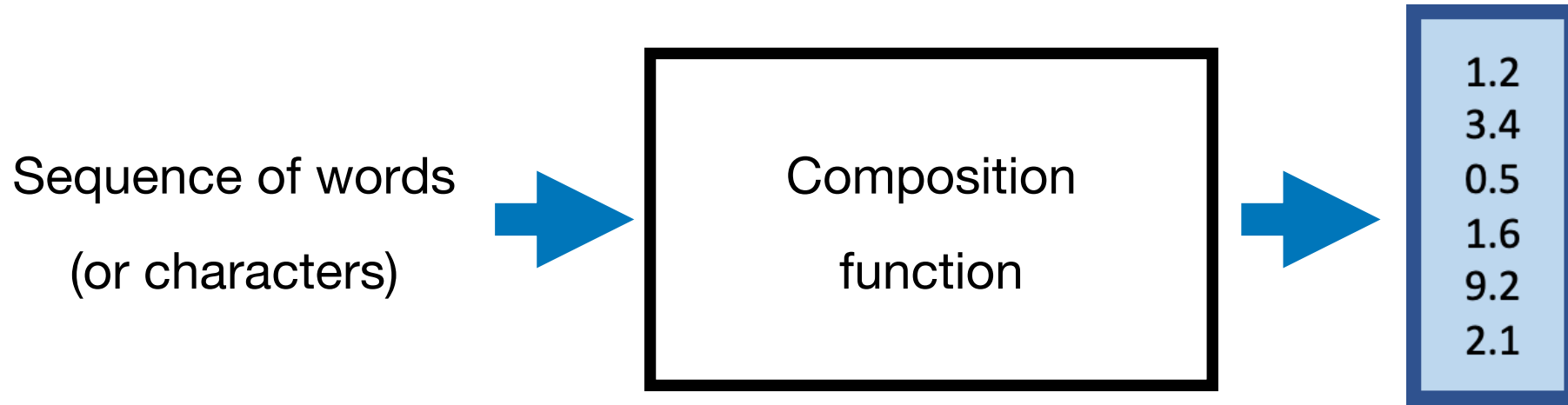
RNN

Transformers

CNN

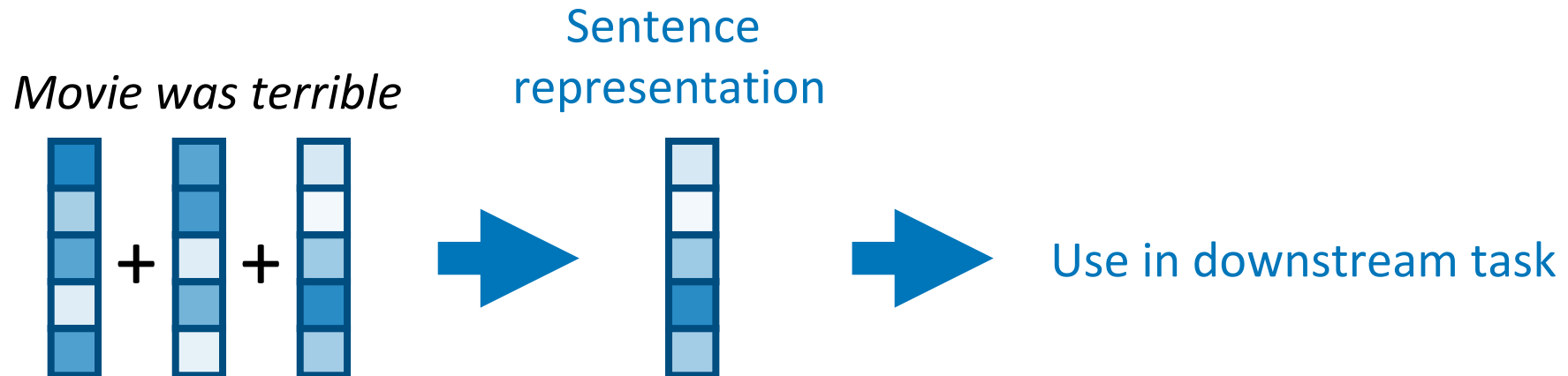
GraphNN

How to **compose** language?



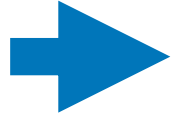
Simplest: take weighted average

Example:



How to use the learned **representation**?

What do we do with this vector representation after we get it?



Use in downstream task

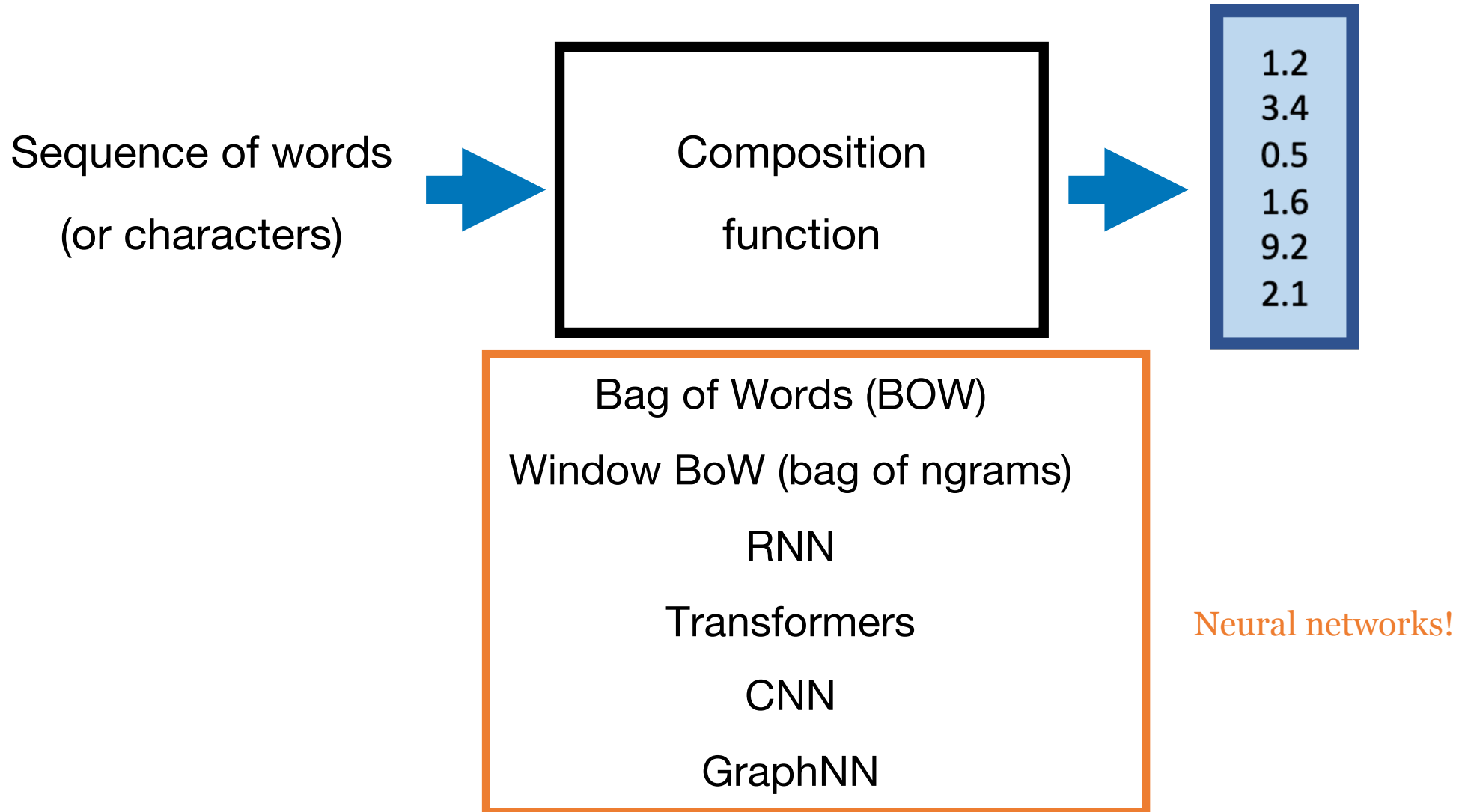
Vector
representation

Do prediction: classify as spam/not spam

Use to generate text (also known as **decoding**)

Can also be used to generate other stuff (like images!)

How to **compose** language?



Thank you!



Getty Images

<https://www.thoughtco.com/syntactic-ambiguity-grammar-1692179>